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A Self-Driving Lab for Battery Electrolyte Design

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Abstract
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As the transition to renewable energy accelerates, development of efficient, safe, and cost-effective energy storage systems become increasingly critical. Lithium-ion batteries dominate the current market, but their technological maturity motivates exploration of alternative chemistries, like aqueous zinc batteries, which offer safety and cost advantages. However, significant challenges remain before such systems can reach practical, industrial deployment. At the research level, battery testing is often time- and resource-intensive, with long testing protocols alongside process variability from factors such as environmental conditions, component quality, and assembly precision. Self-driving labs (coupling automation and data-driven experimentation) offer a pathway to improved reproducibility, higher throughput, and deeper insight into these electrochemical systems. This thesis presents ODACell, a self-driving lab designed for automated coin cell assembly, in-line battery cycling, and closed-loop electrolyte optimization. The initial ODACell prototype is described and evaluated using an aqueous lithium battery model system, revealing poor reproducibility that motivated hardware and workflow improvements: integration of computer vision, environmental enclosure, and increased space efficiency. The upgraded ODACell was applied to multiple electrolyte systems using Bayesian optimization as the experiment planner. Co-solvent effects (from acetonitrile, dimethyl sulfoxide, trimethyl phosphate, and water) on Coulombic efficiency (CE) were explored with ODACell, confirming the detrimental impact of water. In another study with conventional carbonate-based electrolytes, formation cycle CE was mapped for varying additive concentrations (of 1,3,2-dioxathiolane 2,2-dioxide, prop-1-ene-1,3-sultone, tris(trimethylsilyl) phosphite, and vinylene carbonate); Bayesian optimization combined with statistical analysis identified first-cycle benefits from specific additives while confirming known resistance increases from vinylene carbonate. Finally, ODACell was demonstrated on a zincsystem, highlighting its adaptability across chemistries. Optimization of zinc salts ($ZnSO_4$ and $ZnCl_2$) and additives (thiourea, ammonium phosphate monobasic, and sodium dodecyl sulfate) was supported by a separate advanced characterization technique to provide physiochemical explanations to observed CE effects. Overall, this thesis demonstrates the capabilities and challenges of ODACell as a self-driving lab for liquid electrolyte optimization in coin cell batteries.

Keywords: Li-ion batteries, Bayesian optimization, high throughput, self-driving lab, aqueous batteries, automation, electrolyte, additives

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Dedicated to my family

List of papers

This thesis is based on the following papers, which are referred to in the text by their Roman numerals.

I Yik, J. T.; Zhang, L.; Sjölund, J.; Hou, X.; Svensson, P. H.; Edström, K.; Berg, E. J. Automated electrolyte formulation and coin cell assembly for high-throughput lithium-ion battery research. *Digital Discovery* **2023**, *2*, 799–808.

My contribution: I was part of the planning of ODACell; I assembled and programmed the platform. I prepared materials, performed all experiments and conducted analysis. I wrote the manuscript with input from co-authors.

II Yik, J. T.; Hvarfner, C.; Sjölund, J.; Berg, E. J.; Zhang, L. Accelerating aqueous electrolyte design with automated full-cell battery experimentation and Bayesian optimization. *Cell Reports Physical Science* **2025**, *6*, 102548.

My contribution: I took part in the planning of the study with my supervisors. I incorporated Bayesian optimization libraries given by C. Hvarfner into ODACell. I performed all experiments. Analysis of Bayesian optimization data was done with input from C. Hvarfner; analysis of OEMS data was done with my supervisors. I wrote the manuscript with input from co-authors.

III Yik, J. T.; Berg, M.; Thöming, L. L.; Sjölund, J.; Zhang, L; Berg, E. J. Exploring Automated Approach and Data-Driven Insights for Optimization of Electrolyte Additive Blends for Li-ion Batteries. *In Manuscript*.

My contribution: I planned the experimental work with input from my supervisors. I supervised as well as contributed to the preparing and conducting of all experiments. I conducted the data analysis for the Gaussian process model. I wrote the manuscript with input from co-authors.

IV Yik, J. T.; Thöming, L. L.; Vanoppen, V.; Sjölund, J.; Zhang, L; Berg, E. J. Trials and Tribulations of High-Dimensional Electrolyte Design for Aqueous Zinc Batteries Using Bayesian Optimization. *In Manuscript*.

My contribution: I planned, prepared, and performed the experimental work except for the EQCM-D experiments. I analyzed the data associated with the Bayesian optimization, and I contributed to the statistical analysis with alternative models. I wrote the manuscript with input from co-authors.

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List of Abbreviations & Symbols

Table I. *List of general abbreviations.*

CE	Coulombic efficiency
CVA	Coefficient of variation
ESW	Electrochemical stability window
HER	Hydrogen evolution reaction
ML	Machine learning
SEI	Solid electrolyte interphase

Table II. *List of abbreviations for chemicals, electrolytes, and electrodes.*

ACN	Acetonitrile
DMSO	Dimethyl sulfoxide
DTD	1,3,2-dioxathiolane-2,2-dioxide
EC	Ethylene carbonate
EMC	Ethyl methyl carbonate
LFP	$LiFePO_4$
LiTFSI	Lithium bis(trifluoromethanesulfonyl)imide
LP57	1 M $LiPF_6$ in 3:7 vol% EC/EMC
LTO	$Li_4Ti_5O_{12}$
MAP	Ammonium phosphate monobasic, monoammonium phosphate
NMC811	$LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$
PES	Prop-1-ene-1,3-sultone
SDS	Sodium dodecyl sulfate
TMP	Trimethyl phosphate
TMSPi	Tris(trimethylsilyl) phosphite
TU	Thiourea
VC	Vinylene carbonate

Table III. *List of symbols regarding electrochemistry and electrolyte mixing.*

Q	Capacity
c	Concentration
I	Current
ρ	Density
U	Energy
m	Mass
M	Molar mass
R	Resistance
V	Voltage
v	Volume

Table IV. *List of symbols regarding Bayesian optimization.*

α	Acquisition function
$K_\nu(x)$	Bessel function
$K(x,x')$	Covariance function/matrix (cross-correlation in lowercase)
Φ(x)	Cumulative distribution function of the normal distribution
ε	Experimental noise
κ	Exploitation/exploration hyperparameter for upper confidence bound acquisition function
$\Gamma(x)$	Gamma function
$GP(\mu, K(x,x'))$	Gaussian Process
ℓ	Length-scale hyperparameter for radial basis function
μ	Mean
$\mathcal{N}(\mu, \sigma^2)$	Normal distribution
$f(x)$	Objective function
D	Observation dataset
$p(x)$	Probability (distribution)
ν	Smoothness hyperparameter for Matérn kernel
σ^2	Variance

Preface

In compliance with the Policy For the Use of Generative AI in Doctoral Education at the Department of Chemistry - Ångström, Uppsala University, no generative artificial intelligence was used to generate new written content for this doctoral thesis. However, it was used to check grammar and polish certain sentences to improve the flow as well as for language translation. No figures and illustrations presented in this thesis were produced exclusively by generative artificial intelligence; such tools were employed only for identifying Python programming code errors and suggesting minor code improvements.

1. Introduction

In the pursuit of harnessing renewable energy, innovative solutions to store and manage the energy continue to be researched. Moving away from fossil fuels as the primary energy source aims to lower carbon emissions and combat climate change. However, renewable energy sources, such as solar or wind, cannot be conveniently accessed and controlled like fossil fuels, and are typically electrochemically stored. From the International Energy Agency's 2024 World Energy Outlook, between 2010 and 2023, low-emission electricity sources accounted for 57% of the total global electricity generation [1]. Moreover, in the 2025 World Energy Outlook report, growth of renewable electricity generation is projected to increase fourfold by 2050 under the Stated Policies Scenario [2]. The combined pressure from growing energy demands and decarbonization goals pushes research forward to pioneer groundbreaking strategies.

One class of energy storage solutions is rechargeable or secondary batteries. These complex electrochemical systems will be explored throughout this doctoral thesis. Automated experimentation and machine intelligent experimental planning applied to materials engineering can further push scientific understanding of certain battery systems. This chapter first defines the concept of a self-driving lab and outlines its hardware and software components, which together form the foundation upon which accelerated scientific discovery can be achieved. I then introduce the chemistry and associated materials engineering that are explored. Finally, the amalgamation of these elements is framed within the scope of the thesis.

1.1 Experimental acceleration

Within the battery systems, liquid electrolytes often consist of multiple components, creating a high-dimensional design space that complicates the discovery and optimization of novel materials. To efficiently navigate experiments with extensive parameters, automation and machine learning (ML) methods can streamline experimental workflows and accelerate the exploration of electrolyte formulations. The coalescence of automation and ML to accelerate specific tasks or workflows will hereafter be defined as the self-driving lab presented in this thesis.

1.1.1 Lab-scale automation

Industrial automation intended for large-scale production is typically unsuited for research laboratories. They occupy large spaces, are expensive to purchase and operate, and are heavily streamlined for a single purpose. In the small-scale research environment, with limited space and often small quantities (few grams rather than tons) of test materials, automation is focused primarily on a methodology or instrumentation that can aid the researcher through key experimental bottlenecks. This semi-automated experimentation is cheaper, more accessible to academic research groups and more flexible in terms of working alongside researchers and diverse workflows [3].

In the battery field, automation has been developed for the synthesis of solid-state materials, such as electrode materials, or for the characterization of electrodes and electrolytes. For example, the Dahn group published their combinatorial approach for coprecipitation synthesis of lithium-ion cathode materials using a robot for dispensing [4], which the McCalla group then built upon to achieve higher throughput and automated characterization with X-ray diffraction [5]. For liquid electrolytes, characterization is either done directly to obtain properties such as viscosity or ionic conductivity, or by electrochemical testing via cell assembly. Whitacre and Viswanathan *et al.* developed an automated microfluidics platform called Clio to test ionic conductivity of different electrolyte formulations [6]. Matsuda *et al.* developed a robotic experimental workflow that used microplate-based electrochemical cells to test Coulombic efficiency (CE) and overpotential of lithium deposition and dissolution with multi-component electrolyte formulations for Li–O_2 batteries [7]. Automated coin cell assembly allows for electrochemical testing of closed cells, which is sometimes more representative of practical batteries. The Stein group developed their automated coin cell assembly platform, AutoBASS, capable of assembling batches of 64 cells for testing [8]. The Battaglia group also published their Aurora platform, which can automate coin cell assembly, electrolyte formulation, and cell cycling [9]. Commercial solutions are also available, but may require a substantial financial investment and be restricted in workflow flexibility and customization. Diverse and creative automation solutions is continuously being developed. In the thesis, Paper I presents the automated coin cell assembly, electrolyte formulation, and cell cycling platform, ODACell. From Paper II onwards, the platform is integrated with Bayesian optimization, a data-driven ML approach, to efficiently test and optimize the performance of different electrolyte formulations. At that point, the entire platform, dubbed ODACell 2.0, is the self-driving lab presented in this thesis. The platform is shown in Figure 1.1.

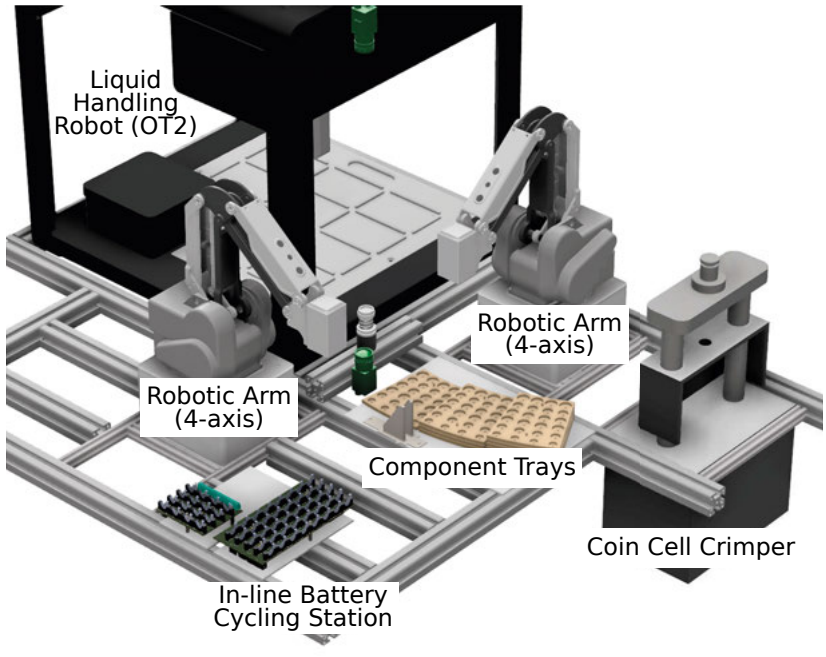


Figure 1.1. Rendering of the ODACell 2.0 platform for automated coin cell assembly and in-line battery testing.

1.1.2 Machine learning

Computations traditionally aid in the calculations of kinetic and thermodynamic properties for complex systems. This has evolved into physics-based and ML models to describe complex systems. Some methods require expensive computational resources, while others require high quantity of data [10]. Regardless, these methods ideally work in tandem with experiments to root the computational predictions in reality. With the advancement of black-box optimization that do not rely on knowledge of the shape of the function *apriori*, direct knowledge from experimental data can be useful for not only optimization, but also developing physicochemical insights.

ML-guided experimentation aims to elevate this insight. Methods such as tree-based models, kernel models, and clustering models can aid battery research [11, 12, 13]. For example, Attarian Shandiz and Gauvin [13] compared the predictive accuracy of multiple models for three category classification using five features of $\mathrm{Li-Si-(Mn,Fe,Co)-O}$ electrode compositions. Deep learning involving neural networks, such as generative models, is also a powerful technique that has shown success [14]. However, these combine computational calculations to maximize information gain. The problem arises from the need for deep learning to use big data, which consists of four major requirements, colloquially referred to as the four V's, namely high volume (large datasets), variety of data, veracity (high-quality), and velocity (high-throughput

and analysis). This is difficult to achieve with current labor-intensive experimental capabilities.

For small datasets and resource- and time-intensive experiments with complex, unknown objective functions, such as battery performance testing, different methods must be used [15]. For example, Shiblee and Laaksonen [16] successfully used a particle swarm optimization algorithm to finetune hyperparameters for an extreme gradient boosting regression model predicting state of health and remaining useful life of large-format batteries. Gaussian process regression models have also been a common tool for predicting state of health and lifetime for large-format batteries [17, 18, 19]. Bayesian optimization is particularly interesting for its advantage of working well with low quantities of data and is powerful for black-box optimization tasks [20]. For this reason, the thesis focuses on using Bayesian optimization to guide experimental workflows. Integrating Bayesian optimization with automation poses a significant advantage to produce high-fidelity data and enables higher throughput, building towards autonomous self-driving labs.

1.1.3 Self-driving labs

The infrastructure of a self-driving lab consists of three major components: (i) the hardware needed to conduct automated experiments, (ii) the software needed to control the hardware as well as analyze the data, and (iii) the experimental planner or machine intelligence to interpret the resulting data and instigate the hardware for subsequent experiments.

The highest-level of self-driving labs can be thought of as the fully autonomous "robotic chemist." There are different levels of the self-driving lab's autonomy depending on the sophistication of the hardware and software. The different levels are detailed by Tom *et al.* [21] and in Figure 1.2. Level 4 self-driving labs exist for robotically "simple" tasks, but Level 5 remains unobtainable in the field [21]. ODACell, as presented in this thesis, resides at Level 4. Below that, Level 2 and 3 self-driving labs encompass the majority of existing platforms. They are characterized by either having fully automated workflows with no machine intelligence (termed automated experimentation, described in Section 1.1.1) or having a robust ML-based software but relying solely on manual experiments (ML-informed experiments, described partly in Section 1.1.2). Level 2 self-driving labs may also have some ML involvement with only certain tasks automated, but not the entire workflow. Level 3 self-driving labs establish the baseline autonomy with significantly decreased reliance on the researcher.

The automated hardware for self-driving labs can be specialized or for general tasks, or combined in the form of open hardware. Specialized hardware include the Opentrons OT2 pipetting robot for liquid handling, while robotic arms are generalized hardware. The integration and in-house creation of

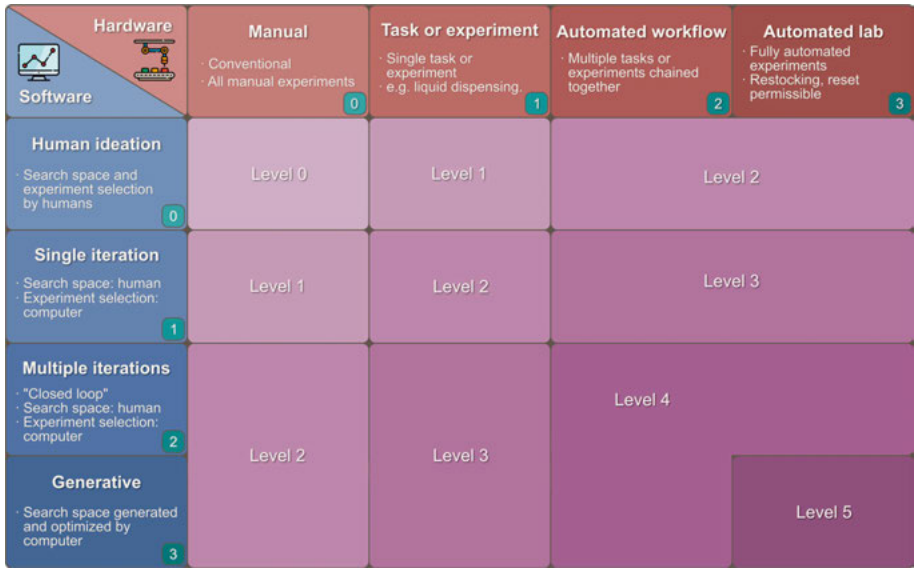


Figure 1.2. Software and hardware autonomy categorization defining the level of autonomy for the self-driving lab. Reproduced from Tom *et al.* [21] Copyright 2024, The Authors, Published by American Chemical Society.

hardware beyond commercially available setups elevate the automation to open hardware. ODACell, presented in Paper I, can be categorized here. The addition of accessory hardware components, such as computer vision or digital twins, elevate the autonomy of the hardware. In the upgraded ODACell 2.0, presented in Paper II, computer vision is implemented.

The software and ML-experimentation planner work in tandem with the hardware and data generated to form the self-driving lab. Orchestration is the interfacing program that keeps track of hardware activity, data management, and the experiment planner all while interacting with the human operator (researcher). Many such software exist and are currently being developed [21, 22, 23, 24]. The orchestration for ODACell was initially built from scratch in Python as an event-loop program for maximum design flexibility. Future developments beyond the scope of this thesis will see the hardware communication protocol be upgraded to align with the SiLA2 standard [25].

The principles and capabilities of experimental acceleration through self-driving labs are now established. Next, the model application domain, rechargeable batteries will be introduced. Within these systems, their chemical complexity, multidimensional design space, and central role in enabling the transition to sustainable energy technologies present a particularly compelling case for accelerated experimentation.

1.2 Rechargeable batteries

A battery is an electrochemical device capable of storing and releasing electricity via redox reactions. Contrary to primary batteries, which undergo irreversible electrochemical reactions, rechargeable batteries can be repeatedly charged and discharged. Thus, for these devices, efficient energy conversion is paramount to provide lasting energy storage. The rechargeable battery consists of a positive electrode, a negative electrode, and the electrolyte. While the electrodes store charge, the electrolyte provides a medium for ion transport and is electronically insulating. The components of the battery are shown in Figure 1.3 and will be introduced in more detail below. The operation of batteries is dependent on a number of interconnected factors involving the electrode materials and electrolyte [26, 27], and their interaction at the interface impacting the electrochemical reactions across it [28]. Cell format and how the cell is assembled also impacts performance [29, 30]. The specific operation and mode of degradation within the batteries are unique to each battery chemistry. This enables batteries to occupy a broad range of applications. Herein specifically, the lithium-ion battery and the aqueous zinc battery systems will be introduced.

1.2.1 Lithium system

Since their commercialization by Sony in 1991, lithium-ion batteries have become a dominant energy storage solution driving global electrification. Figure 1.3 illustrates the operation of a typical lithium full cell[1]: during discharge, lithium ions stored in the negative electrode material is released and shuttle into the positive electrode material through the liquid electrolyte. Simultaneously, electrons are liberated at the negative electrode and travel through an external path to the positive electrode to charge compensate, generating electricity as they conduct through an electric load. At the positive electrode, lithium ions get stored in the structure as the active material undergoes reduction (*e.g.* Co^{4+} to Co^{3+} for $LiCoO_2$). When charging, electricity is used to facilitate the reverse reaction: reduction occurs on the negative electrode and oxidation on the positive electrode, allowing lithium ions to exit the positive electrode structure and enter the negative electrode structure.

The electrode materials determine how much lithium can be stored (capacity, Q) as well as the electric driving force (cell voltage, V), and both contribute to the energy of the cell ($U = \int V dQ$). Capacity and energy are typically normalized with respect to weight (in grams, g) or volume (in liters, L) of the cell and is then referred to as specific or volumetric capacity/energy, respectively. The capacity achieved during discharging (Q_d) relative to the capacity from

[1] A lithium half cell constitutes a working electrode paired with a lithium metal counter and reference electrode to investigate processes on the working electrode without complications from the other electrode, as opposed to a full cell where processes on both electrodes occur and cannot easily be decoupled.

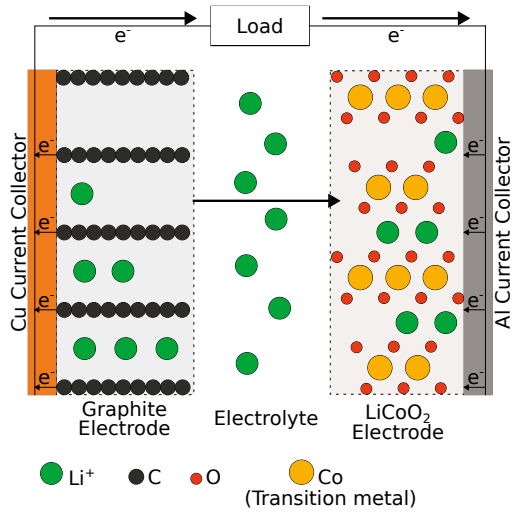


Figure 1.3. Schematic of a rechargeable lithium-ion battery during discharge.

charging (Q_c) is the Coulombic efficiency (CE):

$$\mathrm{CE} = \frac{Q_d}{Q_c}. \tag{1.1}$$

This metric is a prime quantity used throughout the thesis as the performance objective to optimize. Besides the primary redox reaction involving lithium insertion and de-insertion, any other electrochemical reactions will compete and consume charge thereby corresponding to a loss in CE. A beneficial example is the formation of a solid electrolyte interphase (SEI) at the negative electrode, which consumes some of the electrolyte to form a protective layer that prevents further degradation. This process is seen through the low CE in initial cycles, referred to as the formation cycles, before recovering to a steady value, $\gtrsim 99.9\%$; this high CE allows commercial lithium-ion batteries to operate for hundreds of cycles if not surpassing one thousand cycles before reaching end of life [31, 32]. The remaining 0.1% is lost to parasitic reactions including electrode degradation and corrosion, gas evolution, and impurity-related degradation [33].

In addition to CE, other electrochemical parameters can elucidate the internal behavior of batteries. Overvoltage (or equivalently overpotential if using a reference electrode) can be used, such as in Paper III, to indirectly probe changes in resistance. Largely a property attributed to the electrodes, the Nernst equation governs the potential for the redox reaction of batteries, and different conditions and electrode phases result in the characteristic voltage curves seen with specific materials (Figure 1.4) [34]. Under operation (during charging or discharging), additional resistances contribute to the measurable voltage resulting in the voltage hysteresis between charge and discharge. Generally, these are: Ohmic

resistances, following Ohms law ($V = IR$, where I is the current and R is the Ohmic resistance), account for the combined electronic resistance through the materials (*e.g.* current collector, wires, cell casing contacts, *etc.*) and will herein be referred to as internal resistance; charge transfer or interfacial resistance, associated with the kinetics involved at the electrode-electrolyte interface for the electrochemical process; and mass transport or diffusive resistance, attributed primarily to solid-state ion diffusion through the electrode active material based on the porous electrode model [35]. Electrochemical impedance spectroscopy attempts to quantify these resistive contributions; however, specialized instruments are required, which was not a part of ODACell. Instead, other techniques, such as the intermittent current interruption technique [36, 37] applied in Paper III or the galvanostatic/potentiostatic intermittent titration technique can extract internal and diffusive resistances (and measure the diffusion coefficient for solutes in host materials in half-cells under certain assumptions). Different components, or differences in the SEI composition, will then lead to different overvoltages due to contributing different resistances.

Thus, the selection of electrode active materials allow lithium-ion batteries to have a relatively broad range of specific energy (and specific power). The materials undergo redox reactions at certain potentials and the difference between the potentials of the two electrodes is the cell voltage. For cathode materials, $LiNi_xMn_yCo_{1-x-y}O_2$ (NMC) is commonly used, especially the Ni-rich variations ($x > 0.5$), to deliver high specific capacity ($\sim$200 mAh g^{-1}, nominal operating potential >3.6 V *vs.* Li^+/Li) [27]. In comparison, $LiFePO_4$ (LFP), another cathode material, has lower specific capacity (160 mAh g^{-1}, nominal operating potential 3.5 V *vs.* Li^+/Li) but better safety and lower cost [38]. Graphite is conventionally used as the anode material, having a low nominal operating potential of 0.1 V *vs.* Li^+/Li (specific capacity 372 mAh g^{-1}). Conventional carbonate-based electrolyte components get reduced at this low potential, but the SEI prevents further continuous degradation of the electrolyte and enables stable cycling [39]. Another, anode material is lithium titanate, $Li_4Ti_5O_{12}$ (LTO) [40]. The use of LTO as an anode instead of graphite improves the safety of the battery at the cost of performance—the zero-strain structure contributes to high cyclability, and the higher nominal operating potential (1.5 V *vs.* Li^+/Li) compared to graphite results in decreased side reactions on the anode but a lower cell voltage [41]; LTO also has low specific capacity (175 mAh g^{-1} compared to graphite's 372 mAh g^{-1}). There is a wide spectrum of electrode materials besides the four mentioned (and used in the thesis); however, the investigation of electrode materials were unrelated to the research focus.

The electrolyte is a critical component that influences battery performance. It provides a medium for ion transport between electrodes and supports electrochemical reactions at the interface. The commercial lithium-ion battery electrolyte consists of a lithium conducting salt dissolved in a combination of linear and cyclic carbonates, typically $LiPF_6$ in a mixture of ethylene carbonate

(EC) and ethyl methyl carbonate (EMC), diethyl carbonate, or dimethyl carbonate [42]. Through decades of research, these baseline combinations have continuously improved performance, enabling the lithium-ion battery to operate as well as it does today [43]. Besides high conductivity to maintain continuous transport of lithium ions, electrolytes should be chemically and electrochemically stable. The electrochemical stability window (ESW) determines within what potential boundaries the electrolyte is inert. The components will react (either oxidize or reduce at more positive or negative potentials, respectively) outside the ESW. Figure 1.4 shows the electrode operating voltages used in this thesis along with the electrochemical stability windows of aqueous and non-aqueous electrolytes. Aqueous electrolytes, such as those used in lead-acid batteries, are limited by water's ESW of 1.23 V. The choice of the organic carbonate solvents extends this window and provides acceptable chemical and transport properties [43, 42]. On the negative electrode (anode), graphite is still outside the ESW. The gap is covered by the formation of an SEI layer, which passivates the electrode surface and protects the electrolyte from continuous degradation. The fluorinated electrolyte also contributes to the SEI [44] as well as passivates other components inside the cell, like the aluminum current collector on the positive electrode (cathode) that would otherwise continuously corrode [45]. Despite this, research still continues for alternative electrolyte formulations that are safer while providing favorable properties needed for high performing batteries, such as low viscosity, high thermal and chemical stability, environmental benignity, and suitable interfacial properties [42]. Section 1.3 will examine the different approaches.

1.2.2 Zinc system

While lithium-ion batteries have a well-established reputation and has reached a high level of technology readiness, aqueous zinc batteries are emerging as a promising energy storage solution with significant potential for sustainable grid energy storage. Non-aqueous lithium-ion batteries have high production costs and safety concerns from flammable and toxic electrolytes; aqueous zinc batteries using zinc metal anode are cheaper due to abundant resource availability, and aqueous electrolytes for zinc metal batteries are inherently safer and less toxic compared to conventional non-aqueous electrolytes [46, 47]. Zinc metal also has a high theoretical specific capacity (820 mAh/g). The electrodes and electrolyte differ from the lithium system in a few ways. Typically, the zinc metal is the negative electrode, where zinc metal plating and stripping occurs during cycling. On the positive electrode, considerable progress has been made with manganese oxide-based materials, vanadium oxide-based materials, and Prussian blue analogues, spanning different electrode reaction mechanisms [48, 49]. As its name suggests, the aqueous electrolyte uses water

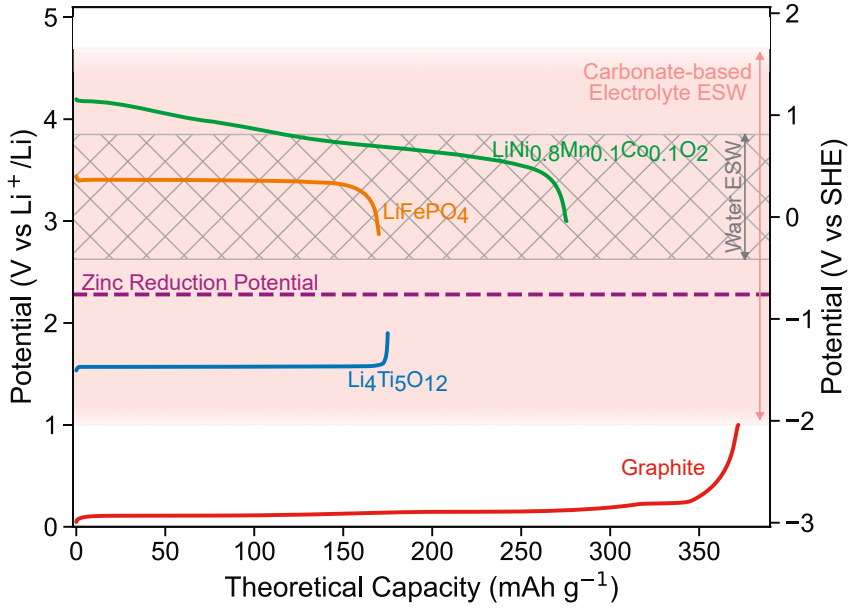


Figure 1.4. Operating voltages of the electrode materials used in the thesis. The shaded region represents the electrochemical stability window (ESW) of the non-aqueous carbonate-based electrolytes, while the hatched region represents water's ESW at pH 7. The electrode curves are relevant for the lithium system, displayed on the left y-axis, while the zinc reduction potential relevant for the zinc system is displayed on the right y-axis (using the standard hydrogen electrode (SHE) reference).

as the solvent with a zinc-containing salt. $ZnSO_4$ is a typical choice of salt, although others such as $ZnCl_2$ have been considered [50, 51, 52].

There are major challenges that aqueous zinc batteries must overcome to reach practical application. Specifically on the zinc metal anode, the zinc standard reduction potential (-0.76 V vs. standard hydrogen electrode) is outside the water stability window, which triggers the hydrogen evolution reaction (HER). The hydrogen evolution not only increases the pressure of the cell, but also changes the local pH around the electrode and can cause associated corrosion [53, 49]. Dendrite formation from heterogeneous nucleation and growth of zinc is another challenge that leads to reduced reversibility, and thus CE, and rapid cell failure from short circuit [47]. To investigate the processes only on the zinc metal electrode, symmetric zinc cells can be used. Symmetric cells have zinc metal for both the positive and negative electrode. Meanwhile, for asymmetric cells, zinc is plated and stripped on a metal substrate instead, which offers some advantages. In symmetric zinc cells, both electrodes provide an effectively unlimited zinc reservoir for repeated plating and stripping, such that cell lifetime and dendrite-induced failure dominate performance assessment. In contrast, asymmetric cells plate and strip a finite amount of zinc per cycle,

making CE a more informative metric for probing immediate parasitic reactions, including the HER, that take capacity away from the electrodeposition and stripping process. To address the adverse effects of dendrite formation and HER, one approach is to explore and alter the electrolyte to increase cycling capacity retention and CE.

1.3 Electrolyte engineering

The electrolyte is a crucial component of any electrochemical cell regardless of chemistry, and so changing the formulation, or properties will have an immediate effect on ion transport and interfacial processes driving the electrochemical reactions. Interfacial reactions are key to both lithium-ion batteries [54] and aqueous zinc batteries [55, 56]. The aforementioned SEI formation in lithium-ion batteries involves the reduction of EC at the electrode surface [57]. For high-voltage electrodes, the operating potential of the cathodes may be outside the anodic stability of the electrolyte. An equivalent cathode electrolyte interphase can be formed to passivate the cathode allowing higher energy densities to be achieved without drastically shortening lifespan [58]. In the zinc system, the electrochemical plating and stripping is a surface process and the properties about the interface directly influences the reactions [59].

1.3.1 Interfacial processes

In brief, at the polarized surface of the electrode, the solvated ions are drawn to the surface. Close to the electrode, at the outer Helmholtz plane, the ion sheds its solvation shell partly to adsorb on the surface of the electrode, occupying the inner Helmholtz plane, to undergo the electrochemical reaction [60]. Therefore, changing the solvent not only changes transport properties like lithium ion transport number, but also changes the solvation energy and the structure of the solvation shell. Altering the salt will also have an influence considering molecular geometry [50, 61]. Additionally, changing concentrations will change the availability of the molecules and affect local attraction and repulsion forces. Already, these parameters—solvent, salt selection and salt concentration—can be finely tuned to engineer an improved battery. The notion of solvent mixtures have already been introduced in Section 1.2.1. Multiple salts and/or multiple solvents are not new, although the complexity of physical interpretation becomes more challenging the more interactions with different components there are.

1.3.2 Co-solvents

Focusing on solvents, the electrolyte can often be categorized as aqueous or non-aqueous in liquid electrolytes. Water provides the fastest ion transport and

inherent safety, but is plagued by a narrow stability window and facilitated corrosion. Non-aqueous electrolytes use organic solvents [62], but also includes ionic liquids [63] and eutectic mixtures [64]. There are also polymer electrolytes [65] and gel electrolytes, moving towards hybrid electrolytes, however they have their own unique challenges. The thesis will be restricted to liquid electrolytes, and the focus of this section will be on co-solvents for aqueous liquid electrolytes.

Research for next-generation lithium-ion batteries includes using alternative salts and solvents, which are safer by being non-flammable, less toxic, and less environmentally harmful. One solution involves moving towards aqueous and water-tolerant solvents and their associated salts. Water is inherently safe and non-flammable, but the narrow ESW and problems associated with proton and solvent co-intercalation [66], challenge its practical use. Insights into water (and the lack thereof) within the electrolyte are the main focus of Paper I and II. The electrochemical stability window can be extended by having a combination of water with other co-solvents in the electrolyte, where the main mechanism attributed to this benefit is altering the solvation structure of the lithium ion. Dimethyl sulfoxide (DMSO) [67], trimethyl phosphate (TMP) [68], and acetonitrile (ACN) [69, 70], used in Paper II, are co-solvents that function to change solvation/desolvation energies for HER suppression. Their structures are shown in Figure 1.5. The solvation structure dynamics are also important to consider as a means for altering electrochemical stability windows and interfacial kinetics [71, 72, 62]. In the highly concentrated

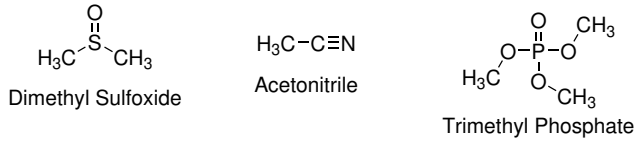


Figure 1.5. Structures of the co-solvents used in addition to water for Paper II.

salt domain, little or no free solvent molecules are present in the electrolyte, reducing their activity. The water-in-salt electrolyte, described by the Xu group [73], achieves a large ESW by increasing the concentration of the salt, lithium bis(trifluoromethanesulfonyl)imide (LiTFSI), to the extreme. In organic solvents, the $TFSI^-$ anion is a good alternative to $PF_6{}^-$ because of its wider thermal stability and similar conductivities [74, 75], while for aqueous electrolytes, LiTFSI has the ability to highly dissociate in water, achieving 21 mol $kg^{-}1$ solvent (molal, m) concentration. However, preparing 21 m electrolyte requires abundant salt (thus high cost and increased toxicological risk), so achieving high performance with dilute salt concentrations is potentially rewarding. In this thesis, only the dilute electrolytes (<3 m) are primarily explored.

While most changes in concentration and primary components have varying moderate effects related to reaction kinetics, another approach for engineering electrolytes involve adding functional molecules called additives in limited amounts (typically <5 % by total weight).

1.3.3 Additives

Electrolyte additives serve many functions within the battery. Regardless of chemistry, additives are generally selected for specific properties intended to obviate or delay a performance-degrading process. For the discussion in this thesis, they can generally be classified into scavengers, SEI formers (or cathode electrode interphase formers if acting on the positive electrode), or electrolyte enhancers. Scavengers target and neutralize particularly harmful substances in the electrolyte. For example, completely eliminating water in the electrolyte is nearly impossible and excess water reacts with the fluorinated salt, which can then degrade the electrode or react with other electrolyte components [62]. Tris(trimethylsilyl) phosphite (TMSPi) is an additive that has been shown to react with $LiPF_6$ and slowly remove fluoride ion, allowing cells to have higher capacity retention for longer cycle life [76]. Additives that contribute to the SEI formation are particularly of interest since the properties of the SEI has impacts on the CE as well as rate capabilities. Cracking and break down of this passivation layer exposes the electrolyte to reduction. Similarly, on the positive electrode, a cathode electrolyte interphase allows the use of high voltage electrode materials (>4.5 V *vs.* Li/Li^+) or higher cutoff voltages for current materials, such as $LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$ (NMC811). Vinylene carbonate (VC) is a well-studied additive that benefits battery performance. Some studies proposed VC as an SEI former by polymerizing and forming a poly(VC) film [77, 78, 62], while others demonstrated its role as a water scavenger, reacting to release CO_2 and forming EC in fluorinated electrolyte [79]. It has also been suggested that VC can facilitate the formation of a cathode electrolyte interphase [80]. It oxidizes at >4.2 V [80, 81], and there is a change in the structure of the electrode surface compared to a base electrolyte. 1,3,2-dioxathiolane-2,2-dioxide (ethylene sulfate, DTD) [82, 83], and prop-1-ene-1,3-sultone (PES) [83, 84] are other potential film-forming additives; TMSPi has been discussed as both a scavenger and cathode electrolyte interphase former [76, 81, 85]. These additives are explored in Paper III (structures shown in Figure 1.6).

In aqueous zinc batteries, additives aimed to enhance plating and stripping on the zinc metal electrode are used to prevent dendrite formation, and/or delay hydrogen evolution and the associated corrosion that follows. One class of additives is surfactants. They are molecules that have both a hydrophobic and a hydrophilic end, which typically form micelles in water. In the electrolyte, they form a barrier on the surface of the electrode where the hydrophobic ends deter

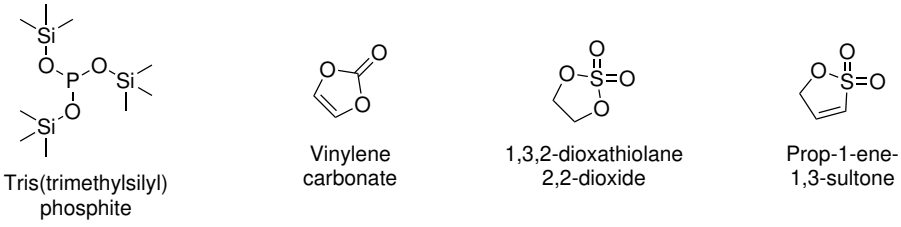


Figure 1.6. Structures of the additives studied in Paper III.

water molecules from interacting with the electrode surface, thus decreasing hydrogen evolution. Sodium docyl sulfate (SDS) is one such additive [86]. In addition to being a surfactant, SDS has also been reported to alter plating and stripping behavior [87], preventing dendrite formation. Thiourea (TU) has been shown to affect the surface of the zinc metal after plating and stripping, leading to increased zinc deposition efficiency [88, 89]. Another additive, ammonium phosphate monobasic (MAP), has been observed to delay hydrogen evolution by altering the local pH at the interface, while enabling smooth zinc deposition [90]. These additives are explored in Paper IV (structures shown in Figure 1.7).

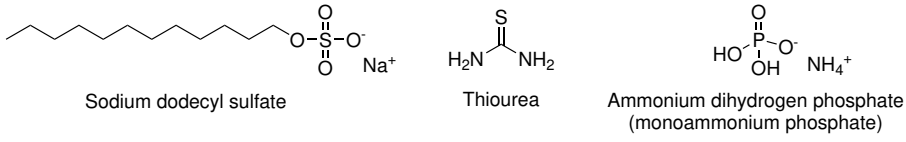


Figure 1.7. Structures of the additives studied in Paper IV.

The selection for electrolyte co-solvents and additives is vast and multi-additive "cocktails" or blends are poorly understood. Additive concentration also impacts the effectiveness of the additives. The large possible search space for effective additives necessitates fast prototyping and intelligent experimental design to accelerate the discovery of novel electrolyte systems. This is what the self-driving lab developed in this thesis aims to accomplish.

1.4 Scope of the thesis

The methodology development towards a self-driving battery lab is the central focus of the thesis. The aim is to demonstrate the feasibility and usefulness of developing an integrated platform where coin cell assembly and the subsequent electrochemical testing via galvanostatic cycling are automated including electrolyte mixing and dispensing, and the experimental design is driven by Bayesian optimization. The scope of the work encompasses the design and implementation of the self-driving lab: the automation system, its integration

with the optimization framework, and the application of the platform to explore additives and electrolyte formulations for lithium-ion and zinc metal batteries.

Paper I focuses on the development of the initial platform, dubbed ODACell, and the performance in a model system outside an inert atmosphere. Uncontrolled environmental factors such as humidity and room temperature prompt discussion on experimental repeatability.

Paper II develops the platform further, encapsulating it in an inert atmosphere (nitrogen glovebox), and presenting the self-driving lab as ODACell version 2.0, with Bayesian optimization as the experimental planning algorithm. Different co-solvents for an aqueous electrolyte were explored as an exploitation-based task for the self-driving lab.

Paper III shifts the focus from aqueous electrolytes to conventional carbonate-based organic electrolytes for lithium-ion batteries. Benchmarking of the new system, conventional NMC811 *vs.* graphite with 1 M $LiPF_6$ in 3:7 vol% ethylene carbonate (EC)/ethyl methyl carbonate (EMC) (LP57), was established before optimization of additives for the formation cycling of the cells were explored. The experimental methodology was developed further with different optimization phases.

Paper IV shifts to aqueous zinc batteries to demonstrate the platform's ability to be applied for diverse applications. Electrolyte additives for increasing zinc metal plating and stripping CE were mapped and general practices for self-driving lab workflows, such as these brought forward, are presented in relation to battery electrolyte screening and optimization.

While the proof-of-concept and model system studies validate the experimental approach, the broader ambition is to illustrate how such platforms can approach diverse next-generation battery systems to accelerate the discovery and optimization of advanced materials. This doctoral thesis aims to present a case study in the development of a self-driving lab for cell-level battery research from the ground up, documenting its iterative improvements and evolution to methodology, hardware, and software. By highlighting practical challenges and subtleties, I hope to provide a foundational reference for other battery scientists and engineers undertaking similar efforts.

2. Methodology

To actualize the automated experimentation with machine intelligence framework for the application of battery assembly and testing, two main constituents of the platform were amalgamated to develop ODACell. The first was the development of the automated cell assembly and coupled in-line testing which provided the physical scaffolding for ODACell. The second was the integration of Bayesian optimization into the programming of ODACell, effectively bridging data analysis and experimental planning. Although these two components can operate independently within ODACell, an orchestration framework was developed to enable their coordinated functionality as well as facilitate a user interface.

2.1 Robotic coin cell assembly and testing platform

The physical aspects of the system can be divided into three parts: the robotic arms for transporting coin cell components; the liquid handling robot responsible for mixing and dispensing electrolyte solutions; and the cycling channels accessible to the robotic arms (first part) and controlled by the orchestration.

2.1.1 Coin cell assembly

Vacuum suction attachment heads were used to pick-and-place the coin cell components. A vacuum head adapter made of polyether ether ketone was inserted into the suction cup of the vacuum head to prevent contamination and degradation of the regular suction cup head. Initially one robotic arm was employed for coin cell component stacking (with two vacuum suction attachment head), but later changed to two robotic arms (one attachment each). This served two key purposes: preventing cross-contamination by assigning one robotic arm to handle cathode-associated components and the other for anode-associated components, and enabling future expansion of the platform through parallel coin cell stacking.

The stacking orientation was dictated by the crimping process, which required the anode casing to face upward. Thus, the stacking proceeded from the bottom: positive electrode casing, positive electrode (cathode), separator, electrolyte dispensing, negative electrode (anode), spacer and spring, and negative electrode casing (with the gasket). In Papers III and IV, an additional

thin spacer (0.5 mm thickness) was inserted between the cathode casing and positive electrode to increase stack pressure. The components, also indicating stacking order, are shown in Figure 2.1. The two most common sources of error from the automated assembly are also shown: component misalignment and electrode deformation. Additionally, a different electrolyte dispensing

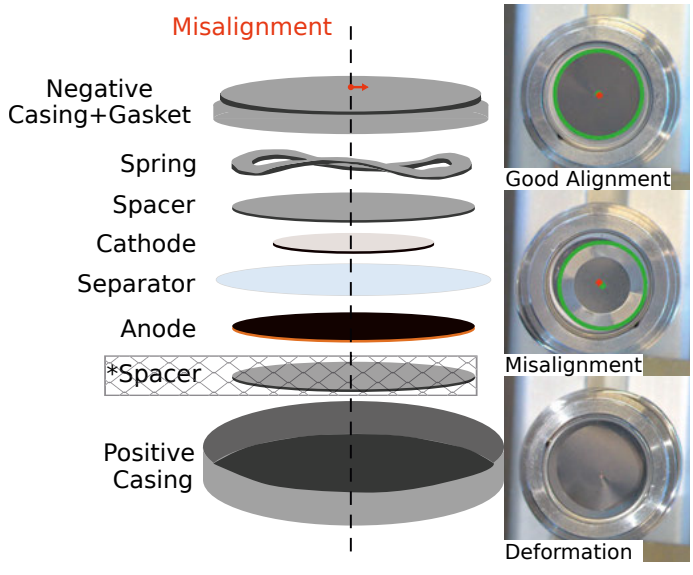


Figure 2.1. Schematic of the coin cell components and how they are stacked during assembly. Misalignment is shown with the red arrow corresponding to the distance of the red and green center points of the images on the right. Images of a well-positioned example (where green and red dots overlap), misaligned example, and an example of an electrode deformity occurring due to the suction of the robot are shown on the right. The boxed, cross-hatched component shows the optional spacer component that is included in Paper III and IV.

procedure was used in Paper III. To assist with wetting and placement, a single droplet of electrolyte was first dispensed between the additional thin spacer and the electrode to flatten and hold the electrode in place by capillary forces. Subsequently, half of the total electrolyte volume was dispensed onto the electrode before placing the separator, with the remaining volume added afterward.

To assist with and record the positional placement of each component, a camera system was installed. Two Basler area-scan cameras (model acA2500-14uc and a2A3840-45ucBAS) took pictures from the bottom (of the component before placement) and the top (of the stack after placement). The positions of the cameras in the platform are seen in Figure 1.1 in green. The center of the component was determined from the bottom picture and used to dynamically offset the placement position. After placement, the top camera records the current stack. Example images are shown in Figure 2.2. A warning system was integrated into the orchestration, where each newly captured image from the

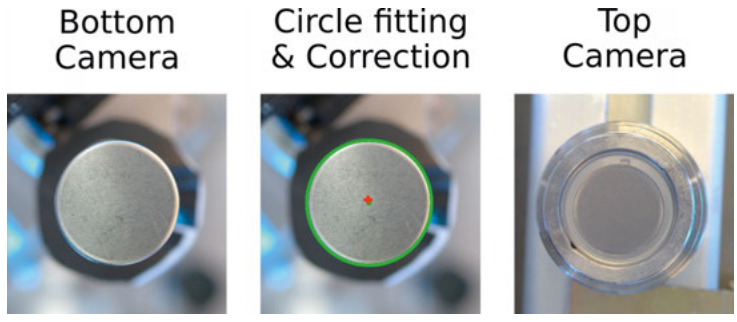


Figure 2.2. Placement sequence of a coin cell component. Component is picked up from the tray and imaged from the bottom (left image); then circle fitting is done to determine and correct offset from optimal center (red dot, center image); the component is placed on the stack and the top camera documents placement (right image).

top camera was compared to the previous one. If the similarity score between the two images exceeded a defined threshold, indicating a potential missing component, the system would pause assembly and query the user to proceed. However, the reliability of this warning system was limited due to manually setting the similarity score threshold to trigger the warning.

2.1.2 Liquid mixing

Commercial options for reliable and high-throughput liquid volume dosing are mature and readily available in the field of pharmaceutics and biochemistry (e.g. Chemspeed [91], Hamilton [92], Mettler Toledo [93]). Opentrons OT2 was procured for liquid mixing—as part of the experimental methodology—and dispensing—as part of the coin cell assembly. The open-source Python application programming interface provided by Opentrons was used to control the robot and was integrated into the orchestration framework via a high-level Python class wrapper.

In the battery field, describing the electrolyte component concentrations can take on different forms. For example, when describing the solvent mixtures, it is typical to use volume ratios (*e.g.* LP57 is 3:7 volume ratio of EC to EMC). Alternatives include stating component ratios by weight or moles, although uncommon. For additives, where the total amount is far less than solvents, this becomes inappropriate. Instead, the amount of additive can be described as the weight percentage of the total electrolyte mass (either including or excluding the additive mass itself). Another way is to just have the concentration in regular molar concentration.

For the studies conducted with the platform it was important to conform to conventional notation for comparison and cross-referencing[1]. However, the OT2 can only be commanded with specific volumes from a source well to a destination well. Thus, different solution mixtures (*i.e.* electrolyte formulations) can only be made by dilutions of stock solutions. For direct interpretation of the Bayesian optimization results, both the Bayesian optimization inputs and outputs were expressed in terms of component concentrations, even though the OT2 operates using volumes. Stock solutions were prepared in appropriate concentration units, and the volumes required to achieve the desired electrolyte formulations were back-calculated accordingly. This makes the Gaussian process posterior model more convenient to interpret, as it is expressed in concentrations rather than raw volumes.

In Paper I and IV, the mixing volumes were simply derived by $c_i v_i = c_f v_f$, where c_i and v_i are the concentration in molar and volume to pipette from the stock solution, respectively, and c_f and v_f are the concentration in the electrolyte formulation and volume of the prepared electrolyte formulation, respectively.

In Paper II, co-solvent amounts were expressed as molar ratio for analysis. However, the parameters used in the Bayesian optimization was volume of each stock solution to simplify the calculations and restrictions. The list of volumes would be directly given to the OT2 to mix, while the molar ratios of each electrolyte component would be calculated. For molality, the usual dilution factors do not apply, but the physical meaning still holds true: the number of moles (or amount of solute) will be the same from one volume, diluted into another volume. The denominator of molal, m, is kilogram of solvent, so

$$c_s m_{s,solv} = c_f m_{f,solv}, \tag{2.1}$$

where c_s and c_f is the concentration (in molal, m) of the starting stock solution and final concentration in the electrolyte for one component, respectively, and m_{solv} is the mass of the solvent. The concentration has the numerator in moles of solute and the denominator is kilogram of solvent. The numerator multiplied by the molar mass of the solute gets the mass of solute (in grams). The adjusted concentration is then the mass of solute over the mass of solvent. This is then simply the mass ratio of solute to solvent in the solution,

$$\frac{\text{kilograms of solute}}{\text{1 kilogram of solvent}} = \frac{0.001 c_s M}{1}, \tag{2.2}$$

where M is the molar mass of the solute. From the ratio, each component can be calculated if a total amount of the stock solution, T, is taken,

$$\frac{A}{B} \to A : B \to \text{amount of } A \text{ from total } T : \quad A = T\left(\frac{A}{A+B}\right) \tag{2.3}$$

[1]Sometimes information in articles are not stated clearly and inferred. However, often, even when all information is present, the stated effects are not necessarily comparable because of different electrodes (or their sizes), cell configuration, or testing protocols.

Since the pipette transfers volumes, the mixing volume v and density ρ can be used to calculate the mass of solute and solvent:

$$\begin{aligned} m_{solute} &= \rho v \left(\frac{c_s M}{1000 \left(\frac{c_s M}{1000} + 1 \right)} \right) \\ m_{solv} &= \rho v \left(\frac{1}{\left(\frac{c_s M}{1000} + 1 \right)} \right) = \rho v \left(\frac{1000}{c_s M + 1000} \right). \end{aligned} \tag{2.4}$$

The factor of 1000 (or 0.001) is to convert grams and kilograms of solute. This assumes homogeneous solution; the sum of components is the amount taken, $\rho v = m_{solute} + m_{solv}$. The final electrolyte volume was a set parameter, so in the Bayesian optimization, the last parameter was removed and a sum constraint was used. Thus, for the analysis and compositional database, the mass was calculated for each component, then converted to moles with their molar mass.

In Paper III, additive concentrations were expressed in weight percentage (wt%) of the total electrolyte formulation mass. Stock solutions were also expressed as weight percentage. Again, the density of the stock solution was needed for the volume calculation as well as setting a total electrolyte mass. An in-depth derivation of the mixing volumes can be found in Paper III Supplemental Information. The dilution from the stock into the electrolyte follows similarly with Equation 2.1, but instead of the mass of just the solvent, it becomes the mass of the entire solution for the amount of stock that is taken, m_s, and the final mass of the electrolyte blend, m_{tot}. The diluted concentration is then the quotient of these masses, scaled by the initial concentration,

$$c_f = c_s \frac{\rho v}{m_{tot}}. \tag{2.5}$$

The volume to transfer, v is then calculated by rearrangement of the equation, and the total electrolyte blend mass m_{tot} is a specified set parameter. In this case, there is an additional well of pure "solvent" (LP57 for Paper III, similar to water without additives in Paper I and II) to facilitate the correct dilution.

2.1.3 In-line battery testing channels and orchestration

To enable automated electrochemical testing after robotic coin cell assembly, the platform needed to include coin cell cycling holders. The challenge was deciding on what holder to implement so that the robotic arms could place the coin cell into the holder and maintain good electrical contact while also having the robotic arms remove the coin cell after testing was finished. Vertical slide-in slotting holders were chosen over horizontal slide-in or push-in from the top based on the ease of a gripper attachment. The coin cell holders were adjusted to allow easier insertion and removal. A printed circuit board was designed and milled to connect the holders to the battery testers. After soldering the

holders and positioning the board to be accessible to the robotic arm gripper, the cycling channels were fixed to the platform. Custom cables were prepared to traverse into the glove box, so that the cycling channels would be on the inside while the electrical equipment was situated outside the glove box.

Initially, 16 cycling channels were implemented. To circumvent the bottleneck of extended testing protocols, the number of channels were increased to 40 with the switch to Neware BTS 4008Tn-5V10mA battery cyclers. Additional 120 channels have been purchased and will be integrated into the system. In this thesis, for optimization campaigns, batches of four or eight cells were assembled at once. Batches were staggered so the model always had incoming data to update the model when querying for additional trials. Channel monitoring triggered a flag for channels with finished protocols, where the cycling data could then be automatically processed and the optimization objective extracted to add into the model dataset. The robotic arm then removed the cell and freed the channel for additional cells.

The specific cycling protocol is determined beforehand and is not changed throughout the optimization. This is all then handled by the orchestrator program. Specifically, each study in this thesis used a different cycling protocol:

- In Paper I, a simple constant current charge and discharge of 0.5 C-rate was used for ten cycles (cutoff voltages 1.6 V-2.2 V).
- In Paper II, constant current charge and discharge of 0.5 C-rate was used for five cycles (cutoff voltages 1.6 V-2.3 V) with a voltage hold at 1.6 V for the final cycle until 1/50 discharge current was reached; then a final cycle at 0.2 C-rate was used to calculate performance metrics.
- In Paper III, formation cycling was done with a constant current dis/charge of 0.1 C-rate (cutoff voltage 3.0 V-4.2 V for benchmarking, 3.0 V-4.4 V for additive optimization) for three cycles. For the benchmarking, no voltage hold was applied; however, for the additive optimization, a voltage hold at 4.4 V during the charge was applied until 1/50 of the current was reached. Then, a 0.5 C-rate was used for 200 cycles. Intermittent current interruption technique [36, 37] was implemented during the formation cycling to extract internal resistance; every five minutes of dis/charging, a five-second rest period followed until cutoff voltages are reached. Internal resistance was then obtained by extrapolation fitting of the rest period and plotted against the state of charge.
- In Paper IV, constant current density plating and stripping of 1.0 mA cm^{-2} was used for 20 cycles.

Orchestration of the ODACell platform is done in Python[2]. A process map of the self-driving lab workflow is shown in Figure 2.3. Low-level application programming interface for the robotic arms and the OT2 liquid mixing robot was provided by the manufacturer. Python class wrappers were constructed to easily manipulate basic functions, such as moving the robotic arm from a

[2]https://github.com/jyik/ODACell

single position to another position. Periphery electronics such as activating the crimping machine or the vacuum of the robot vacuum heads were controlled through the robotic arm digital in/out ports. The high-level orchestration is an event-loop interface. Additional files keep track of changing statuses of the platform such as which tray to take components from or which cycling channels are free or occupied. A separate slave computer handles control of the cycling equipment and protocols, data extraction and analysis, and hosts the Bayesian optimization programming. The event-loop interface on the slave computer creates a worker thread to execute the two types of inputs: commands and queries. The commands run functions that require no return message or information to the master orchestrator computer; queries return some message or information to the orchestrator, such as when querying the Bayesian optimization for the next recommended trial to sample. The master computer event-loop interfaces with the user and takes commands, which also create worker threads to execute them. In this framework, the closed-loop electrolyte formulation, coin cell assembly and testing, and Bayesian optimization incorporation can be done sequentially, while allowing possibility of parallel task completion and workflows.

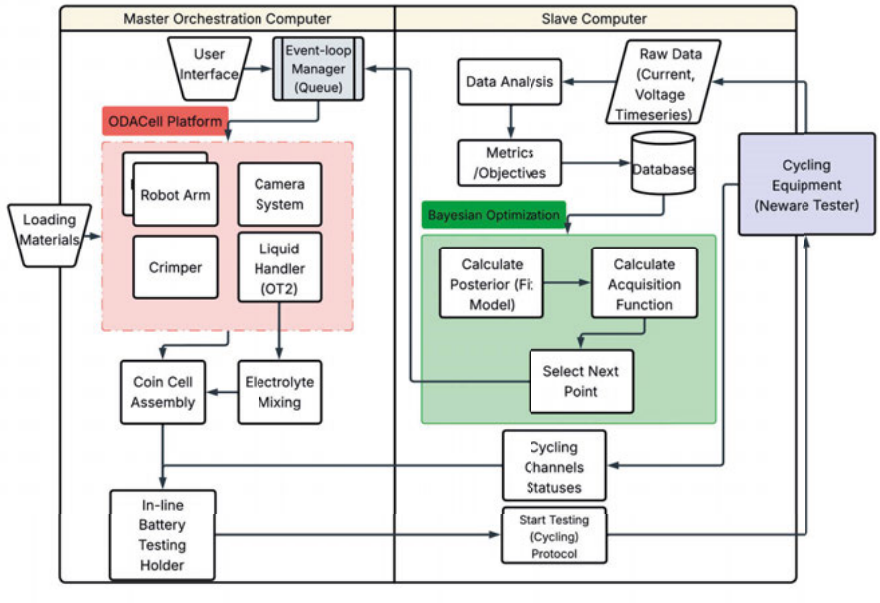


Figure 2.3. Schematic of the orchestration illustrating how the electrolyte screening with coin cell assembly and testing experimentation integrates with Bayesian optimization to form the closed-loop methodology.

2.1.4 Advanced characterization

Expansion of the workflow aims to eventually incorporate advanced characterizations. The ODACell platform mainly sets the foundation for rapid screening, but to understand the fundamental working mechanism that leads to performance improvement, complementary advanced characterizations is necessary. The advanced characterization conducted as part of the thesis were aside from any self-driving lab methodology. These experiments were manual additions to further understand and explain the results found from the Bayesian optimization experiments.

Paper I included Karl Fischer titration technique to quantify the water content of the DMSO solvent. The specifics are irrelevant for the scope of the thesis, but commercial instruments have already incorporated automation for faster and efficient experiments, such as the Metrohm 851 Titrando (Switzerland) used in the paper. In Paper II, online electrochemical mass spectrometry was used for detecting hydrogen evolution. The intermittent sampling variant used in the paper sampled the gas from the headspace of a custom cell while retaining cell pressure [94]. The mass channel associated with hydrogen gas was monitored and the rate of hydrogen evolution was calculated for different electrolyte compositions. This was then averaged across the first three cycles. Thus, the reaction order in the system was explored to determine significant interaction effects. Paper IV used electrochemical quartz-crystal microbalance to understand the detrimental effects of the TU additive. Electrochemical quartz crystal microbalance sensing has been used in electrodeposition and stripping applications to extract viscoelastic and morphological details such as the degree of porosity and thickness of the electrodeposits [95]. In Paper IV, electrochemical quartz crystal microbalance with dissipation monitoring is used to quantify the ratio of charge directed toward zinc deposition versus irreversible passivation or parasitic reactions. Correlating frequency changes with cumulative charge shows how specific additives influence the CE and structural reversibility of the plating/stripping cycles.

The advanced characterization techniques provided complementary, and quantitative insights that helped understand the trends observed with the Bayesian optimization process. In future work, integrating selected characterization techniques for additional metrics could further enhance or expand the optimization framework.

2.2 Bayesian optimization

Optimization is fundamentally a set of iterative decisions to change parameters in order to achieve the best outcome. The basic sequence of steps is listed in Algorithm 1. In this context, the policy applied on the dataset to choose the next sample point can vary, from random sampling to complex algorithms and calculations; regardless, it is used to map the available (known) data, D, to the

location of where to sample next, $x \in \mathbb{X}$, where $\mathbb{X}$ is the domain of the objective function, f. The evaluation of the proposed point on the objective function is denoted as an observation on x to include the consideration of the "black box" optimization. The nature of the objective function, f, may not take on an explicit expression and the decision-making process becomes increasingly difficult as this is made under uncertainty. This is the basis for probabilistic modeling, where we aim to find a model $p(y|x)$ that describes the probability of observing $f(x) = y$ given input x. The Bayesian approach then lends itself to update prior beliefs by incorporating observed data through Bayesian inference.

Algorithm 1 Sequential optimization adapted from Garnett [20] Copyright 2023, Roman Garnett.

Require: initial dataset D ▷ could be empty
  repeat
    $x \leftarrow \text{POLICY}(D)$ ▷ choose next sample location
    $y \leftarrow \text{OBSERVE}(x)$ ▷ evaluate at location
    $D \leftarrow D \cup \{(x, y)\}$ ▷ update dataset
  until resources exhausted
  return D

Bayesian inference involves drawing conclusions about the behavior of the objective from combining both prior beliefs incorporated into a prior distribution and the observational data, $D = \{(x_i, y_i)\}_{i=1}^{n}$. To do this, Bayes Theorem is used to calculate the posterior, defined as

$$p(f|D) = \frac{p(f) \cdot p(D|f)}{p(D)}, \tag{2.6}$$

where

- $p(f)$ is the prior and represents initial beliefs about the objective f before observing any data,
- $p(D|f)$ is the likelihood and represents the probability of observing the data given the objective, f,
- $p(f|D)$ is the posterior and represents the revised probability about f after incorporating the observed data, D, and
- $p(D)$ is the marginal likelihood and represents a normalization factor so the posterior can integrate to 1, computed as $p(D) = \int p(D|f)p(f)df$.

On each optimization iteration, the posterior is then recalculated after the data D is updated. The posterior not only provides a point estimate, but also quantifies the uncertainty around these estimates, which can be used for further computations. The use of an acquisition function takes this posterior and decides the next point to sample.

In the next sections, Gaussian processes and the choice of acquisitions functions will be discussed in as much detail as is required for understanding

the thesis discussion. Optimization stops upon the researcher's satisfaction or the exhaustion of resources.

2.2.1 Gaussian processes

If the underlying analytic form of an objective function cannot be determined or is too complicated to derive, Gaussian processes enable modeling by treating the function as a distribution over possible behaviors [96, 97]. This allows functions to be modeled with varying complexity, not restricted by a fixed set of parameters. This makes Gaussian processes useful for Bayesian optimization as surrogate models.

A Gaussian process is a collection of random variables. Formally, a Gaussian process on f is denoted [20]:

$$f \sim GP(\mu(\mathbf{x}), K(\mathbf{x}, \mathbf{x}')), \tag{2.7}$$

defined by a mean function $\mu(\mathbf{x}) = \mathbb{E}[f(\mathbf{x})]$, and covariance function, or the kernel, $K(\mathbf{x}, \mathbf{x}') = \text{cov}[f(\mathbf{x}), f(\mathbf{x}')]$, which determines how changes in the objective function value at x influence values at other points, x'. This effectively controls the function's structural properties encoding our beliefs such as smoothness and periodicity. In the finite-dimensional marginal distribution, any finite subset follows a multivariate Gaussian (normal) distribution, so a single objective function value is normally distributed,

$$p(f(x)|x) \sim \mathcal{N}(\mu(x), K(x,x)), \tag{2.8}$$

with the mean $\mu = \mu(x)$ and variance $\sigma^2 = K(x,x)$. This property of the Gaussian process inherently allows us to render the typical 95% credible interval, $\mu \pm 1.96\sigma_{\bar{x}}$.

Together, the mean function and covariance function make up the prior distribution. One of the simplest and most used collection of covariance functions is the squared exponential function, also called the radial basis function:

$$K(x,x') = \sigma^2 \exp\left(-\frac{||x-x'||^2}{2\ell^2}\right), \tag{2.9}$$

where the variance σ controls the overall magnitude of uncertainty in the function values, ℓ is the length-scale hyperparameter controlling the smoothness of the function, and $||x-x'||^2$ is the Euclidean distance between the two points. Because this kernel is infinitely differentiable, it is well-suited for modeling smooth functions. Another commonly used family of covariance functions, which was the one used exclusively for this thesis, is the Matérn kernel [98],

$$K(x,x') = \sigma^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu}||x-x'||}{\ell}\right)^{\nu} K_\nu \left(\frac{\sqrt{2\nu}||x-x'||}{\ell}\right). \tag{2.10}$$

The hyperparameter ν controls the smoothness of the function in a much greater capacity than ℓ since the kernel is $\lceil \nu \rceil - 1$ times continuously differentiable. $\Gamma(\cdot)$ is the Gamma function and $K_\nu(\cdot)$ is the modified Bessel function. At the limit $\nu \to \infty$, the Matérn kernel converges to the radial basis function. In this thesis, $\nu = 5/2$ was used. Other kernels exist [99, 100], such as the periodic kernel for oscillating function, and can be combined to encode complex, exotic behaviors; however, the discussion is outside the scope of this thesis. Moreover, there is a diversity of mean functions [20], but only the constant-mean prior was used here. As the name suggests, the constant-mean function is a scalar value (typically set to 0 initially), which is learned from the observed data when computing the posterior. The mean function is important for extrapolation and controls the central tendency of the objective function [20].

Adjusting the standard kernels to include automatic relevance determination [100] enables the model to interpret the impact of each input dimension independently. For each input dimension, a separate length-scale hyperparameter is assigned. The parameter sensitivity or feature importance is then inversely proportional to the length-scale.

To compute the Gaussian process posterior distribution, we condition the Gaussian process prior with n observational data, $D = \{x_i, y_i\}_{i=1}^n$, $y_i = f(x_i) + \varepsilon$ where the observational noise is Gaussian, $\varepsilon \sim \mathcal{N}(0, \sigma_n^2)$. The computation, based on both the prior and likelihood of the data, yields a Gaussian process posterior mean and variance at new test points $\mathbf{x}_*$ as:

$$\mu(\mathbf{x}_*) = k(\mathbf{x}_*, \mathbf{X})[K(\mathbf{X},\mathbf{X}) + \sigma_n^2 I]^{-1}\mathbf{y}, \tag{2.11}$$

$$\sigma^2(\mathbf{x}_*) = K(\mathbf{x}_*, \mathbf{x}_*) - k(\mathbf{x}_*, \mathbf{X})[K(\mathbf{X},\mathbf{X}) + \sigma_n^2 I]^{-1}k(\mathbf{X}, \mathbf{x}_*), \tag{2.12}$$

where

- $\mathbf{X} = [x_1, x_2, \ldots, x_n]^\top$ is the observed input data ("training x"),
- $\mathbf{y} = [y_1, y_2, \ldots, x_n]^\top$ is the observed output data ("training y"),
- $K(\mathbf{X}, \mathbf{X})$ is the covariance matrix of the observed input data,
- $k(\mathbf{x}_*, \mathbf{X})$ is the cross-covariance matrix between the test points $\mathbf{x}_*$ and the observed training inputs $\mathbf{X}$, where $k(\mathbf{X}, \mathbf{x}_*) = k(\mathbf{x}_*, \mathbf{X})^\top$,
- $K(\mathbf{x}_*, \mathbf{x}_*)$ is the covariance matrix of the test points (the covariance of the prior),
- and I is the identity matrix.

This calculation is derived from the joint distribution of the observed data and test points, which also remain Gaussian [20, 100]:

$$p\left(\begin{bmatrix}\mathbf{y} \\ f(\mathbf{x}_*)\end{bmatrix}\right) \sim \mathcal{N}\left(\begin{bmatrix}\mu_D \\ \mu(x)\end{bmatrix}, \begin{bmatrix}K(\mathbf{X},\mathbf{X}) + \sigma^2 I & k(\mathbf{X}, \mathbf{x}_*) \\ k(\mathbf{x}_*, \mathbf{X}) & K(\mathbf{x}_*, \mathbf{x}_*)\end{bmatrix}\right), \tag{2.13}$$

where μ_D is the observation data mean. We can intuitively understand, in simplified terms, the posterior variance as the prior variance subtracted by the uncertainty explained by the observation data. Similarly, the posterior mean can be viewed as a distance-weighted addition to the prior mean (set to 0).

The kernel hyperparameters greatly shape the Gaussian process behavior, so learning or updating those hyperparameters can impact the performance and success of Bayesian optimization [101, 102]. Hyperparameters such as the Gaussian noise, σ_n^2 and length-scales ℓ_i can be updated by different methods. Typically, one method is by maximizing the marginal likelihood of the observed data using gradient-based techniques [20]. The complete treatment of the derivation and computation is omitted here as it is beyond the scope of the thesis, but I note that this process is an inherent part of each Bayesian optimization iteration.

2.2.2 Acquisition function

A central aspect of Bayesian optimization, the acquisition function uses the Gaussian process posterior (or surrogate model) to determine the next location to evaluate within the domain. This is the optimization step that largely influences the optimization balance between exploration and exploitation. Exploration aims to query points in uncertain regions, while exploitation aims to query points that are likely to yield high objective values.

Formally, an acquisition function, denoted $\alpha(\mathbf{x}, D)$ assigns a score to each point in the domain reflecting our optimization preferences based on the surrogate model constructed with the observed data, D. The next candidate, $\mathbf{x}_n$, is determined by maximizing the acquisition function:

$$\mathbf{x}_n = \arg\max_{\mathbf{x}\in\mathbb{X}} \alpha(\mathbf{x}, D). \tag{2.14}$$

While many different acquisition functions exist [20, 103], only a curated selection will be presented here.

Probability of improvement only considers the probability that the objective function value of the next observation is higher than the value of the current observed highest value, $f(\mathbf{x}^+)$:

$$\alpha_{PI}(\mathbf{x}) = p(f(\mathbf{x}) \geq f(\mathbf{x}^+)) = \Phi\left(\frac{\mu(\mathbf{x}) - f(\mathbf{x}^+)}{\sigma}\right) \tag{2.15}$$

where $\mu(\mathbf{x})$ is the Gaussian posterior mean at $\mathbf{x}$, and $\Phi(\cdot)$ is the cumulative distribution function of the normal probability density function since $f(x)$ is normally distributed. Extending this further because probability of improvement does not consider the magnitude of improvement, the expected improvement acquisition function considers both and is among the most commonly used acquisition functions [20]:

$$\alpha_{EI}(\mathbf{x}) = \mathbb{E}[\max(0, f(\mathbf{x}) - f(\mathbf{x}^+))]. \tag{2.16}$$

The class of acquisition functions using the logarithmic version of the expected improvement [104] was used in Paper III. The two acquisition functions

presented above are in the exploitation category. Balancing exploitation and exploration, the upper confidence bound [105] is a simple and effective acquisition function to do this,

$$\alpha_{UCB}(\mathbf{x}) = \mu(\mathbf{x}) + \kappa\sigma(\mathbf{x}), \tag{2.17}$$

where κ is a hyperparameter that controls the balance (higher κ for more exploration, lower κ for more exploitation). A distance-weighted adjustment to expected improvement, that was used in Paper II called diffusion-weighted probability of improvement, increases the diversity of sampling for locating objective function local maxima. This acquisition function provided some exploration and is given by

$$\alpha(\mathbf{x};\mathbf{X}^+) = \min_{\mathbf{x}'\in\mathbf{X}^+} ||\mathbf{x}-\mathbf{x}'||^2 \cdot \mathbb{E}[\mathrm{g}[f(\mathbf{x}) \geq r]], \tag{2.18}$$

where $\mathbf{X}^+ = \mathbf{X}_r \cup \mathbf{X}_{\text{curr}}$ is the set of queried candidates that surpassed the reference value r with the current points under evaluation, $||\mathbf{x}-\mathbf{x}'||^2$ is the Euclidean distance, and $\mathrm{g}[\cdot]$ is a utility function, similar to probability of improvement, to score values larger than the set reference value r.

For exploration, the main acquisition function used in this thesis, specifically in Paper III and IV, is the negative integrated posterior variance [106, 107]

$$\alpha_{\text{NIPV}}(\mathbf{x}) = -\int_{\mathbb{X}} \mathrm{Var}[f(\mathbf{x})|D_x]dx, \tag{2.19}$$

where $D_x = D \cup (x)$ is the observed data with the query candidate point. The negative integrated posterior variance function calculates the average global variance of the posterior if the point was to be evaluated and added to the observation data. This effectively implies the next candidate point $\mathbf{x}_n$ is selected by its ability to achieve the largest decrease in global variance.

Other more complex acquisition functions, such as entropy search [108], as well as the family of versatile Monte Carlo acquisition functions [109] will not be discussed.

2.2.3 Implementation

The Adaptive Experimentation Platform, Ax [110], provided the high-level application programming interface for interacting with the orchestrator and operator. Ax leverages the Bayesian optimization algorithms implemented in BoTorch [106] to create a flexible server-client optimization workflow. Every Bayesian optimization experiment initialized a client with predefined parameters and their search space domains. The choice of acquisition function and number of Sobol initialization points were also initiated with the client. Multiple clients can be created at once and optimized in parallel. The choice of which client to have the orchestrator access was then up to the operator. A client was accessed only when completing the trial or querying the next trial.

In Paper II, three clients were created. The first two were initiated identically with different seeds, each with 40 Sobol points. Both clients used the diffusion-weighted probability of improvement acquisition function, Eq. 2.18, for the 60 optimization trials, each sequentially. The third client served as a Gaussian process container, where the data from the first two clients were imported into the third client and just used to generate the posterior for analysis.

In Paper III, two clients were created. The optimization was separated into three phases, exploration, exploitation, and incorporation of manual data. The first client was initialized with the negative integrated posterior variance acquisition function and eight Sobol random sampling points. After a set amount exploration (100 trials), the next client was initiated with the variant of logarithm expected improvement accounting for noisy observations (and no Sobol sampling). Data from the previous client were imported and the optimization based on exploitation started. After 16 trials, manual decision-making was used instead of the acquisition function to query the domain edge. For clarification, *manual* data constitutes manual decision of the next query point, while the assembly and electrolyte preparation was still conducted by ODACell.

In Paper IV, a single client was used. It was initiated with eight Sobol points, before continuing with 36 trials from the negative integrated posterior variance acquisition function. Instead of allowing only Bayesian optimization trials, manual data was imported into the client to create a more robust posterior, as with Paper III.

The implementation of the clients require creating *experiments* inside the clients. This holds information about the search space domain, acquisition function and other metadata needed for the Bayesian optimization algorithm. Just changing the acquisition function or domain would require creating a new experiment with the desired change and copying over the other metadata into the new experiment and replacing the existing experiment; this is because only a single experiment can be present within a client. This is the reason for the new clients to be created and the data manually added and completed in the new client.

2.3 Statistical analysis

Aside from Gaussian process modeling to fit the data, other methods were employed. Pearson correlation was calculated in Paper II (parameters and objectives) and III (component placement and objective) to get a correlation matrix. The Pearson statistic gives a value between -1 and 1 providing the direction and degree of linearity (magnitude 1 indicating ideal linear fit). The advantage of Pearson correlation over calculating the feature importance of the Gaussian process model is its simplicity and easy interpretation. Possible co-dependence in parameters may also be deduced with co-linearity. The dis-

advantage of this method is also its simplicity; higher order or non-monotonic relationships in the data are not captured and may therefore be underestimated or misinterpreted.

Ridge regression was used in Paper III and IV as another form of correlating parameters and the objective(s) [111]. Ridge regression also provides a linear predictive fit, but with increased sensitivity to co-dependence among parameters. It introduces an L2 regularization term, which penalizes large model coefficients, thereby mitigating overfitting and multi co-linearity. The same disadvantages with higher order or non-monotonic relationships apply with this method.

Other modeling approaches, such as random forest regression [112] and SHAP-based analysis [113], were evaluated, but were excluded from the results and discussion because they either yielded trends consistent with those already observed or demonstrated limited global predictive accuracy in cross-validation, reducing confidence in their interpretability.

3. Results and discussion

3.1 Establishment of lab automation

Design to automate the coin cell assembly process began with unrestricted bounds. The culmination of the first working prototype ODACell is presented in Paper I. My design choices and reasoning for why components were chosen are explained in detail below. Then, its first practical application is discussed.

Three four-axis robotic arms (DOBOT MG400, China) were scattered across a 150 cm by 200 cm working bench so that the entire space was accessible by at least one robotic arm (schematic of the layout illustrated in Paper I). Robotic arms provide a versatile tool to manipulate objects with the use of different attachment heads. The essence of coin cell assembly is component stacking, so the robotic arms were suited for the repetitive task. Vacuum suction attachment heads (SMC Corporation) were chosen to move and stack components, allowing direct vertical placement. The robotic arm attachments are shown in Figure 3.1.

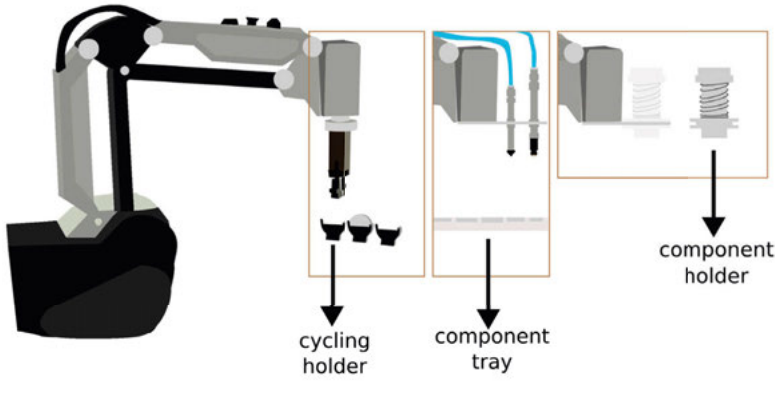


Figure 3.1. Illustration of the three robotic arm attachments for the first ODACell prototype: gripper (left), vacuum suction (middle), and claw (right). ODACell 2.0 removed one robotic arm and split the vacuum suction attachments between the other two robots for dual functionality.

To simplify the stacking, component trays were designed. Complicated processes such as conveyor-belt pick-up or mechanical spring-loaded stacks were considered but abandoned at this early stage for simple hard-coded pick-and-place locations with predefined locations for each component. The simplest tray design would be to order the component slots in the order of pick-and-place; however, this would mean the vacuum suction head would be in contact

with the active material on one electrode and then contact the other electrode (after picking up and placing the separator). A cleaning process for cross-contamination seemed unnecessary, so a secondary suction head attachment was introduced instead (Figure 3.1, middle). Since both vacuum heads were attached to the same robotic arm, the component trays were designed to avoid cross-contamination. The first three components: positive casing, electrode, and separator was placed in order on the tray; the next three components were placed in reverse order on the tray: electrode, spring and spacer, negative casing (Figure 3.2). The stacking location, referred to hereafter as the working area, was fixed and placed between two robotic arms: the robot with the vacuum suction heads and the robot with the claw to lift the holder where the stack was. Because the two vacuum head positions were fixed to the same robotic arm and unable to change orientation, rotating the arm would result in a radial trajectory, hence the radial distribution of component lines on the trays.

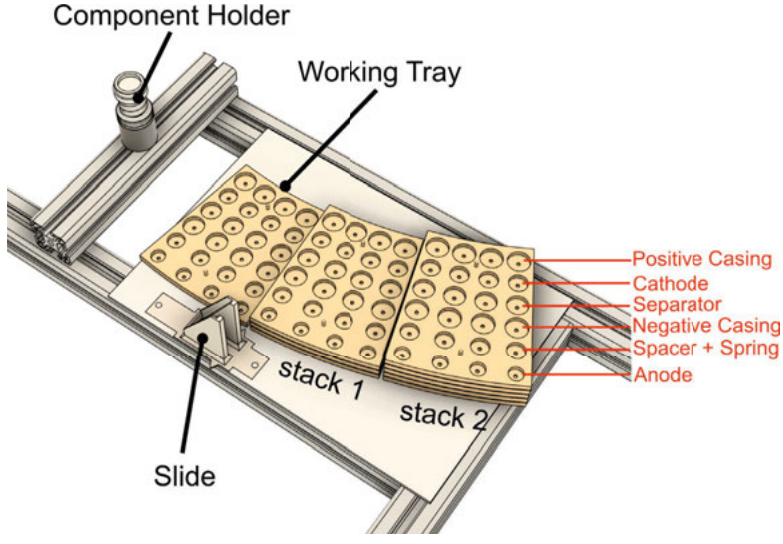


Figure 3.2. Rendering of the component trays, component holder, and slide from ODACell 2.0. The working tray and additional trays in stack 1 and 2 are shown. These core components were inherited from the first prototype of ODACell.

Since the pick-up locations were hard-coded for one tray at one location, that position was fixed as the working tray. Four cells (columns) were milled onto each component tray. Positions beside the working tray were also created and fixed to stack multiple trays. The tray working sequence was established: the stacking process would occur for each of the four columns of the tray, and once the tray was empty, the vacuum suction head would remove the tray and take a new loaded tray off the adjacent stacked component trays onto the working tray location and continue (see the visualization in Figure 3.2). The robot with the claw attachment was responsible for bringing the entire stack to

different locations (Figure 3.1, right). The first prominent location was within the liquid handling robot. Instead of a separate attachment for liquid handling, a commercial pipetting robot, Opentrons OT2, was purchased and repurposed for liquid electrolyte mixing and dispensing. Two main wellplates were used in addition to the virtual "wellplate" where the robotic arm and cell stack holder interacted with the liquid handling robot: stock reservoirs plate and mixing electrolytes plate. Between the two stacking processes, the robot with the claw brings the cell holder to the liquid handling robot to dispense the electrolyte.

Different mixing procedures were considered. The standard in biological laboratories are using vortex vibrational mixing. At the time of ODACell's purchase and development, a vortex mixing attachment was infeasible and expensive. Instead, pipette mixing was used since no solid dosing was considered. Different parameters can be adjusted for pipette mixing including location of mixing within the well, speed of aspiration and dispensing, volume of mixing, and number of aspiration-dispensing cycles. Only speed, volume, and number of cycles were adjusted. The location remained unchanged—at the bottom of the well. The volume of mixing was adjusted to 700 µL, and then changed to 80% of the electrolyte volume in Papers III and IV. The aspiration/dispensing speed was adjusted to the maximum speed of the OT2 Python protocol, 300% the normal speed, to encourage turbulent mixing.

The number of mixing cycles was carefully considered. Because of miscibility differences between compounds, no single number of mixing cycles were used and so was adjusted for each application. Advanced techniques were considered and attempted in order to ensure homogeneous solution after mixing, namely infrared spectroscopy, but development was postponed due to time constraints. The number of mixing cycles were chosen based on visual inspection. In the initial experiments reported in Paper I, DMSO-water mixtures appeared visually homogeneous after approximately 10 mixing cycles (Water and DMSO are highly miscible). Nevertheless, the mixing protocol was extended to 20 cycles. Paper II and Paper IV used 30 mixing cycles.

The second prominent location for the robot with the claw attachment was to bring the finished stack into the crimper to close the cell. Digital input and output signals controlled the triggering of the crimper and when the crimping was finished. A reverse movement sequence was used to bring the closed cell in the cell holder back to the working area for the robot with the suction attachment to pick up the cell. In-line battery cycling provided the means to test the assembled cell immediately after assembly. The holders for battery cycling were important to the workflow. The holders needed to secure the coin cell, while being able to be inserted and removed by the robotic arm. Many holder designs exist and can be categorized into vertical and horizontal cycling positions (some examples shown in Figure 3.3). The horizontal cycling positions were convenient as the robot with the vacuum suction could push the coin cells in directly. However, difficulties arose when attempting to remove the cell. The clamps were too tight for the suction cups to remove the coin cell

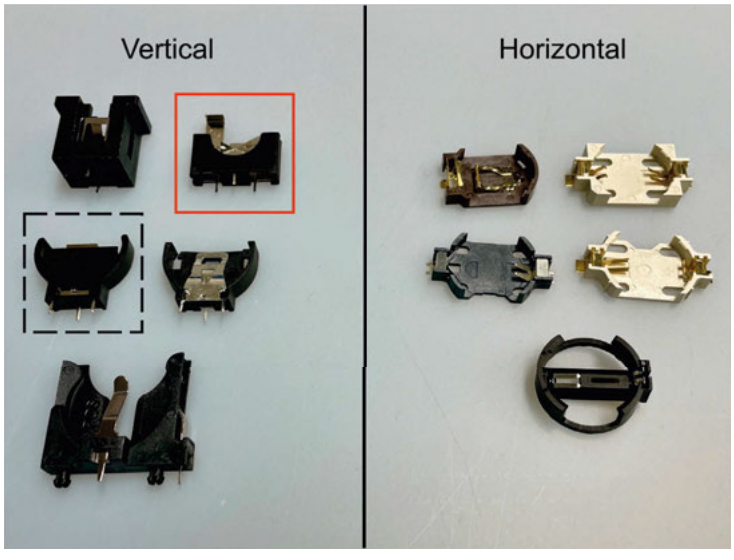


Figure 3.3. Examples of commercially available coin cell holders for electrochemical testing, separated into vertical or horizontal (if holder width corresponded to the coin cell thickness or diameter, respectively). The dashed box represents the coin cell cycling holder selected for the first prototype of ODACell; the red solid box shows the coin cell cycling holder selected for ODACell 2.0.

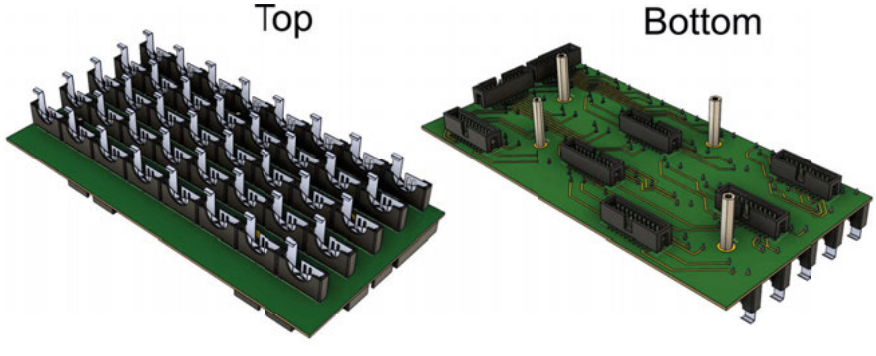


Figure 3.4. Printed circuit board design for the cycling station of ODACell 2.0. Each board held 40 cell cycling holders. The first prototype of ODACell only had 16 channels.

from the holder. When the clamps were loosened, it became hard to secure the coin cells to the electronic contacts. Horizontal slide-in holders were also too tight and incompatible with the vacuum suction heads. Therefore, vertical holders were used.

The third four-axis robotic arm was fitted with a gripper attachment (Figure 3.1, left). The gripper switched between closed and open states via a digital

input signal. The vertical position cycling holders were accessible with the gripper attachment. Multiple different commercially available designs were tested, until one was decided upon. After snapping off a security clamp on the holder (top metal piece of the holder in Figure 3.4), the coin cell fit well into the holder and was easily removed by the robot with the gripper attachment. A custom printed circuit board design increased space efficiency (Figure 3.4). Thus, the cycling station, consisting of the custom circuit board, wire connections to the battery cycler, and the cell cycling holders were added to the automated workflow. A 3D printed slide was designed to convert the horizontal coin cell to the vertical position (Figure 3.2).

3.1.1 Reproducibility in an uncontrolled environment

After all the assembly sequences were connected together via the orchestration program, the first tests were assembling 83 LFP||LTO full-cells with 2.0 m $LiClO_4$ in DMSO electrolyte. The system was chosen for ease of handling and

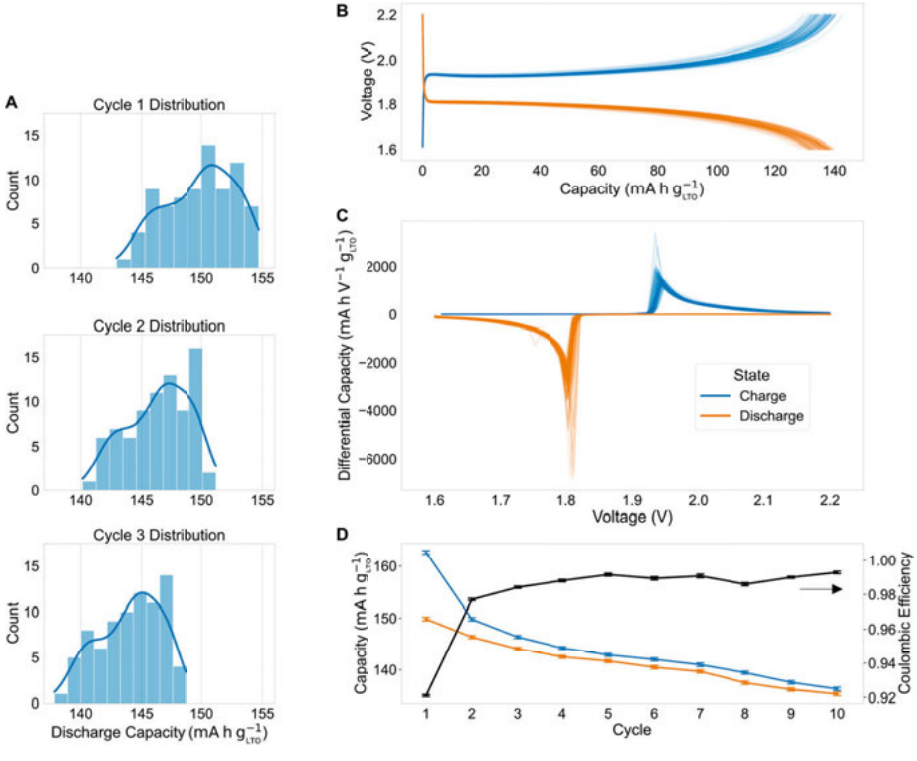


Figure 3.5. Reproducibility of 80 cells assembled by the first prototype of ODACell. A) Distribution of discharge capacity across three cycles indicates consistent distribution across cycles. B) Charge and discharge curves and differential capacity plots C) overlaid indicate no obvious outliers. D) Cycling progression indicates stable cycling with progressive degradation. Error bars are standard error.

simplicity. The LFP electrode was selected because its operating potential lies within the ESW of water, eliminating concerns related to oxygen evolution. LTO was chosen instead of graphite due to its less reductive potential, which helps avoids excessive hydrogen evolution—a DMSO-based electrolyte ensured that both electrode potentials remained within the ESW of DMSO. Under these conditions, hydrogen evolution at the LTO electrode arising from water contamination was expected to be the primary contributor to performance degradation. Reproducibility of the model system in the laboratory open atmosphere is observed in Figure 3.5. The histogram indicated a deviation from normality (seemingly a possible bimodal distribution). At this early stage of ODACell, the detrimental effects involving water contamination and HER were considered more substantial than misalignment (resulting in three failed cells) and small variations in the electrode mass loading. These observations further motivated the need for reproducible experimental conditions. Batch-wise assembly exposed the system to different ambient conditions (*i.e.* morning or afternoon), and the repeated usage of a stock electrolyte likely contributed to increased data scatter as more water was increasingly absorbed into the organic solvent-based electrolyte after sequential uses. Figure 3.6 illustrates the

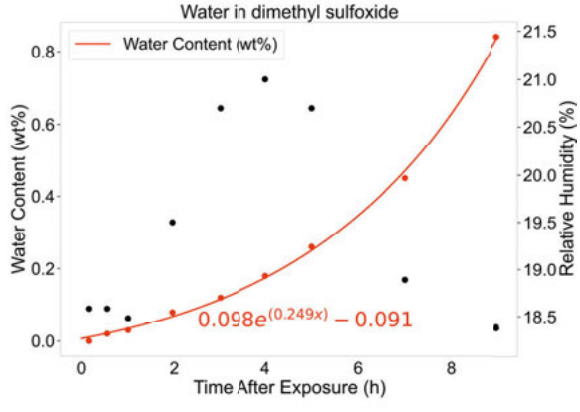


Figure 3.6. Water content in only dimethyl sulfoxide solvent after exposing it to ambient laboratory environment near the liquid handling robot, OT2. The red dots represent measured water amount in the dimethyl sulfoxide solvent after open atmosphere exposure in the morning (9:00). The black dots represent the humidity reading at the OT2 before each water content measurement. The temperature of the room was controlled by central heating at 19.7 ± 0.1 °C.

hygroscopic nature of the electrolyte, showing increasing water content during open-air exposure as well as unstable humidity conditions (higher humidity at midday).

A water series was conducted to determine the relationship between water content and its associated degradation in the DMSO system. Five levels of

water-DMSO binary mixtures were tested with 12 repeats. All 12 cells were assembled right after preparing the electrolyte to ensure similar conditions among groups. Figure 3.7 shows noticeable decrease in capacity starting at 8 vol% water. Distinct for 16 vol%, the spread in discharge capacities were exacerbated with increasing cycles.

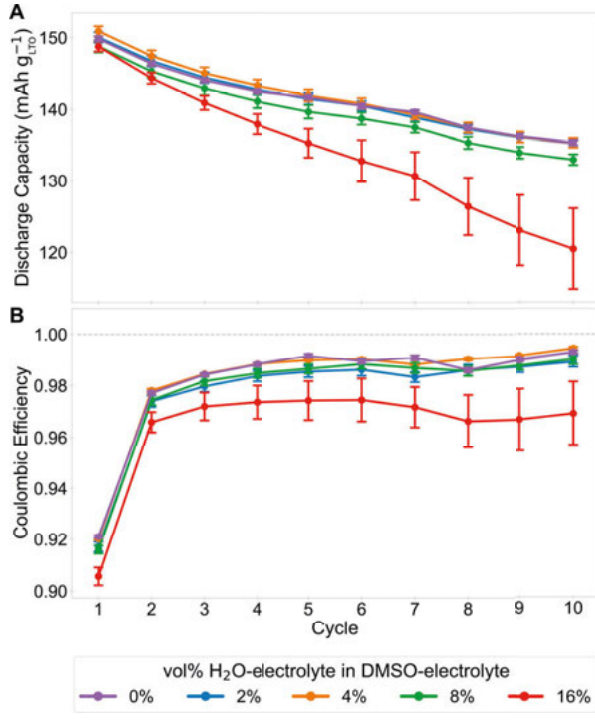


Figure 3.7. Performance degradation of different binary mixtures of water and dimethyl sulfoxide (DMSO). A) Discharge capacity and B) Coulombic efficiency are consistent. Error bars are standard error.

This study demonstrated several important angles to present the data. Unlike traditional methods, comparing absolute response objectives of single replicates, herein, the mean and spread (as standard error here) are considered. Each aspect provides insights into the system as well as inspire confidence in the interpretations drawn. Better transparency is highlighted that may be useful for the community by presenting and discussing outliers or failed cells (*e.g.* via misalignment).

Ultimately, the performance (capacity and CE) and reliability lending to research usefulness was too limited having the ODACell platform operate without controlled atmosphere. To be able to incorporate machine learning methods and intelligent decision-making as well as expanding explorable chemistries, the platform had to be moved into an inert environment. Nevertheless, these

simply binary DMSO-water mixtures were used to confirm mixing protocol and provide a foundation for more complex workflows.

3.2 First steps of ODACell

Paper II chronicles two major evolutions: the implementation of Bayesian optimization, effectively bringing machine intelligence to the experimentation creating a self-driving lab; and encasing ODACell in an inert atmosphere. The latter will be discussed in greater detail in the next section.

The first application of the self-driving lab was the exploration and optimization of the co-solvents design space for aqueous electrolytes. The implemented experimental planner was Bayesian optimization, where I, the experimentalist, determined the domain and other set boundaries; only the dilute salt concentration in a sufficiently aqueous electrolyte (>20% molar ratio of water) was explored by setting water amount as an additional objective. The results and discussion around Paper II will be presented here. Based on the experimental data points and ranking them to determine patterns among several high performing samples, the results showed two regions for high CE, one with high amounts of DMSO, and the other with high amounts of TMP (Figure 3.8). Performance here considers the tradeoff between CE and discharge capacity,

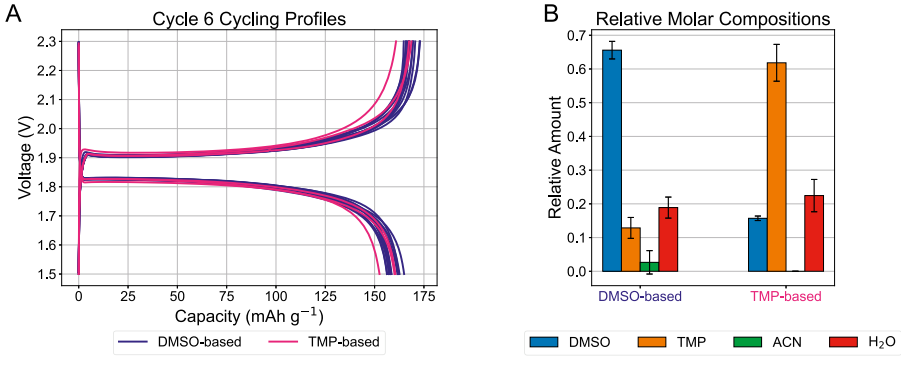


Figure 3.8. Sorted highest Coulombic efficiency (CE) of sixteen electrolyte formulations revealed high molar ratios of dimethyl sulfoxide (DMSO) or trimethyl phosphate (TMP). A) Charge and discharge curves of the two groups show similar CE. B) Relative molar compositions for each group show low amounts of acetonitrile (ACN), and water around the desired minimum, 0.2. Error bars are standard deviation.

a multi-objective optimization. The optimization progression with objective tradeoff is shown in Figure 3.9. The multi-objective optimization adds extra dimensionality and can be interesting to observe the limits of each objective in relation to each other. However, for Paper II, CE and discharge capacity have a strong correlation with each other, so the response is similar (resulting in a

linear relationship on the tradeoff plot). This system would likely converge to a point in the infinite iterations limit instead of a visual Pareto-front.

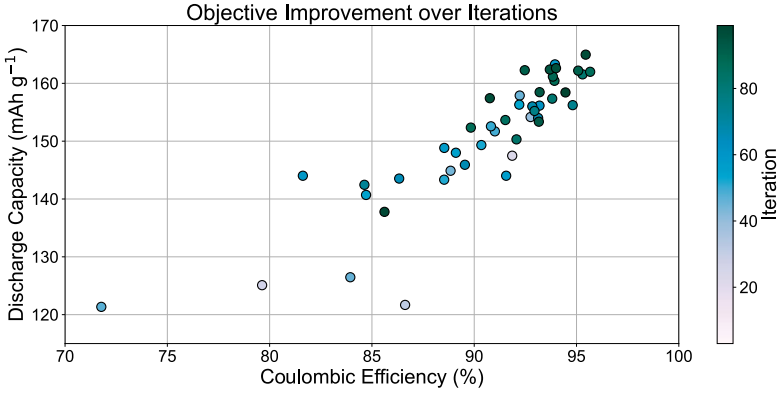


Figure 3.9. Coulombic efficency and discharge capacity tradeoff throughout the optimization progression. As the number of optimization iterations increase, the closer the points are to achieving the Pareto-optimal points.

The multi-dimensionality of the optimization task complicated the analysis. Correlations presented in the study (Pearson correlation) assumed independence and a monotonously increasing or decreasing trend across the domain. Contour maps provided more insights and escaped the oversimplifications of bivariate correlations. Specific concentration ranges responsible for high performance could be visually seen (Figure 3.10). However, the visualization is still limited; the contour maps are still two dimensional slice projections of the entire parameter space such that the other parameters must be set to a fixed value. Overlaying experimental (training) data on top risks giving a false sense of areas sampled on the reduced dimensional space. In some cases, the natural choice is to set the other parameters to zero, effectively removing them to explore isolated interactions. However, this could affect model interpretation greatly if the uncertainty in that region is higher (due to less frequent sampling in those regions, in this case). This can result in poor model validation metrics for interpreting global trends and predictions. Reverting to simpler methods, such as linear regression or other models, discussed in the next section with presentation of Paper III, may inspire more confidence in drawing global conclusions. The choice of how to represent the Bayesian optimization findings and meaningfully interpret the results has been a conscious challenge throughout the works.

Repeats of random Bayesian optimization-derived electrolyte formulations were conducted and shown in Figure 3.11. This practice was carried throughout the thesis. It provides confidence in the results obtained for the model beyond the variance derived from the prior and data. In Paper II, the CE repeatability was compared against their electrolyte composition. Although no statistically significant correlations were found between the standard deviation and compo-

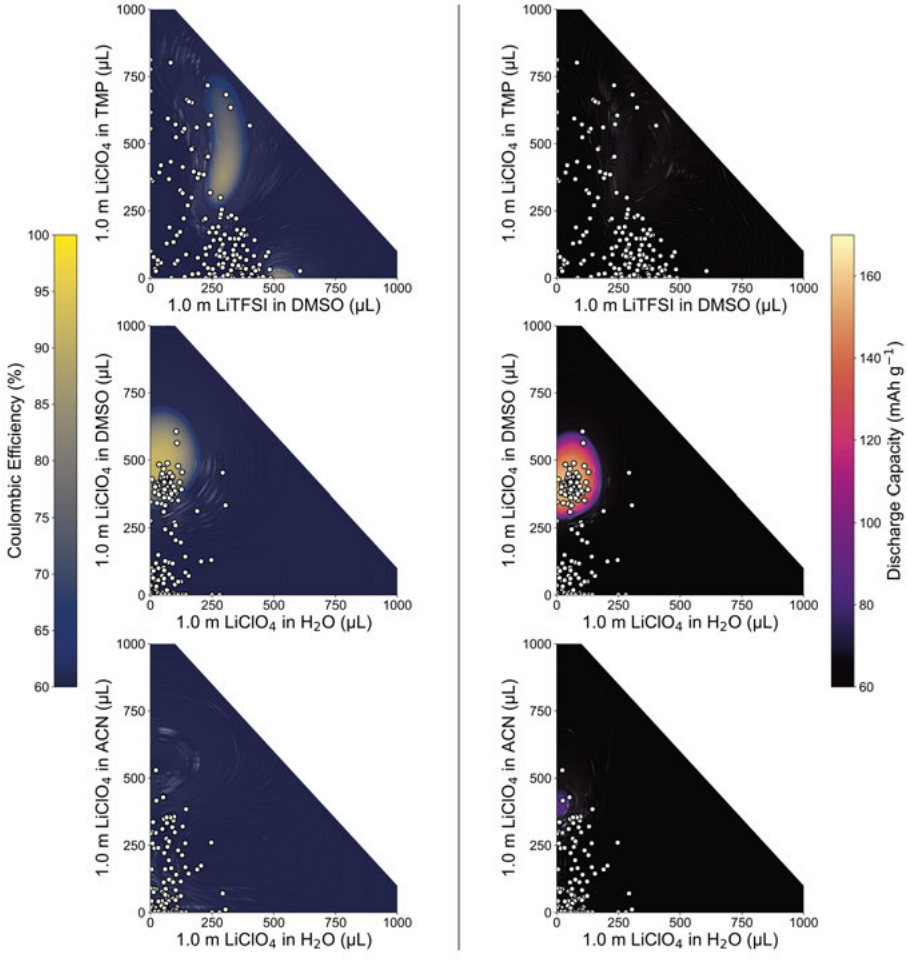


Figure 3.10. Two dimensional contour maps of the Gaussian Process posterior showing three representations to highlight different objective responses: two maxima (top row), one maximum (middle row), and no maxima/response (bottom row). Coulombic efficiency objective (left) and discharge capacity objective (right) underscore the correlation between the multi-objectives.

sition (extremely small sample size), at least one trial (Trial 80) showed a larger variance. The electrochemical side reactions (primarily hydrogen evolution reaction) could be contributing to the variance as explained in Paper II, but it challenges the homoscedasticity assumption. The heteroscedastic treatment in the posterior computation (Equation 2.11 and 2.12) would be to allow the variance σ_n^2 be a matrix/function instead of a constant variable. The consequence of this would be changing uncertainty (credible interval) around certain regions despite sampling. In practice, minor heteroscedasticity would translate to posterior uncertainty being underestimated in regions of larger experimental

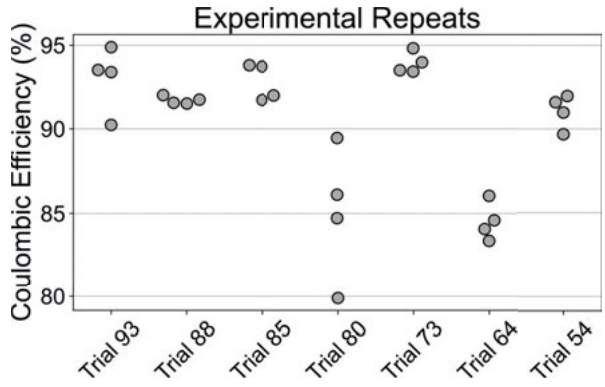


Figure 3.11. Experimental repeats of different trials throughout the optimization show possible heteroscedasticity.

variance. Significant heteroscedasticity may alter the posterior model and, thus, conclusions drawn from the model. Unfortunately, without knowing the nature or deriving a function for the heteroscedastic behavior, it was difficult to attempt to explore this implementation, and I assumed that the impact was only minor, thus keeping a narrow focus on facile implementation. Instead, inclusion of the repeats into the Bayesian optimization dataset gave a better informed posterior mean and variance around those regions.

Although the Bayesian optimization methodology was successful in finding high-performing electrolyte formulations, removal of water was not permitted in the experimental design, and the salt concentrations were not explored. The questions remains: what would happen if the water concentration was not biased and the salt concentration was also explored? That is to say, if the Bayesian optimization had unrestricted bounds. In this case, the two extrema are considered: one where the water was eliminated (the source of the major degradation mechanism is removed), and one where the salt concentration increased to enter the water-in-salt electrolyte regime. Considering the salts of the system, the LiTFSI water-in-salt electrolyte ESW is only cathodically stable until $\sim$1.9 V *vs.* Li^+/Li [73]; therefore, the water-in-salt electrolyte would not eliminate the HER completely (LTO electrode operates at 1.5 V *vs.* Li^+/Li). The solubility will also be a challenge with the co-solvents, as it is unclear if a water-in-salt type electrolyte would be possible. Moreover, with high concentration and mixed co-solvents, diminished Li^+ transport (conductivity) may also negatively impact performance. Thus, there is only a slight possibility of finding maxima in the water-in-salt regime (situation involves conditioning all variables) compared to just removing water (situation involves conditioning a single variable). The possibility of maxima between these two extrema depends on whether there would be an synergetic interactions with the components and in what concentrations. This was the region Paper II hoped to operate in. In general, the behavior of the optimization progression depends on the

acquisition function. With limited resources, exploration would be difficult to fully map the response, but could provide an indication of better performing regions to explore (*i.e.* tending to go lower water concentration or higher salt concentration). From the sampling behavior I observed, maxima on the parameter domain extrema are less sampled, so more optimization iterations would be needed. For exploitation, the large "distance" between the maxima would likely result in the optimization missing local maxima. The initial dataset seems to influence the early optimization progression and which maxima direction it follows. Throughout the thesis, Sobol sampling was used, which provided an unbiased starting point for the optimization. To fully understand the optimization behavior with unrestricted salt concentrations and extended co-solvent bounds would require performing the proposed experimental campaign; the fine tuning of the optimization settings would impact its direction and optimization success.

3.2.1 Advanced characterization: gas analysis

Knowing the primary degradation of the system, hydrogen evolution, advanced characteristics in the form of gas analysis was employed to better understand

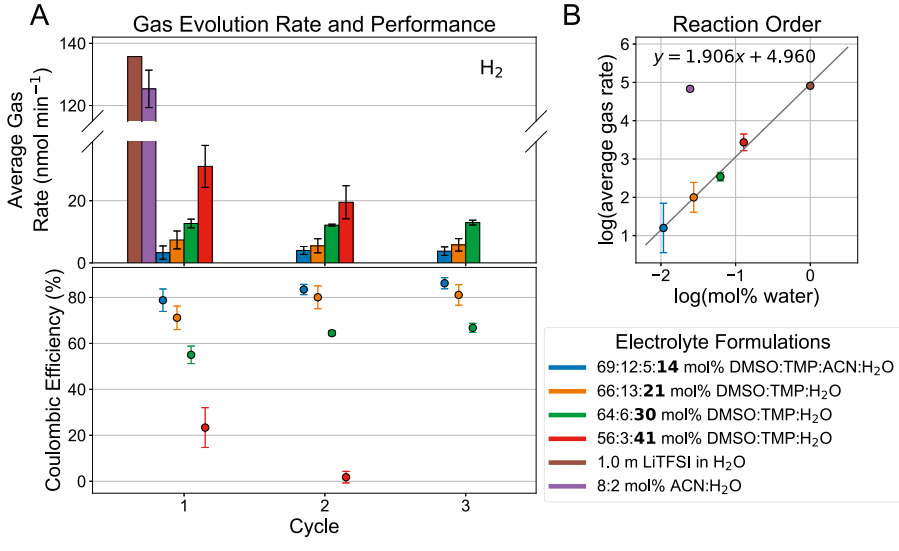


Figure 3.12. Average hydrogen evolution rate and reaction rate for different electrolyte formulations containing different water content. A) The average gas evolution for the first three cycles. Higher hydrogen evolution rate negatively impacts the Coulombic efficiency. B) Logarithm-logarithm plot of the gas rate and water concentration established expected reaction kinetics that the acetonitrile (ACN)-based electrolyte deviated from. Error bars are standard deviation. DMSO, dimethyl sulfoxide; TMP, trimethyl phosphate.

the effects of the electrolyte formulation on the performance degradation. The average hydrogen gas evolution rate was contrasted against the CE for different electrolyte formulations (containing different amounts of water content) in Figure 3.12. As expected, higher average gas evolution rates coincided with lower CE, reflecting charge diversion from lithium insertion to the HER. The relationship between water content and gas evolution was modeled with a power-law fit. This relation was motivated by first-order reaction kinetics. While other functional forms were applied, the resulting power-law offered the best balance between fit quality and physical interpretability. From this, I concluded ACN with water (4:1 molar ratio) increased the HER rate beyond what would be considered expected based on the water content. This provided a better understanding for why the Bayesian optimization did not find any maxima with ACN (Figure 3.10 bottom row). The workflow integrating rapid screening with subsequent advanced characterization proved to be effective.

3.3 Refinement of ODACell

3.3.1 ODACell 2.0

Experiments conducted for Paper II, III, and IV occurred within an inert atmosphere. To enclose ODACell, adjustments had to be made to the platform to increase space efficiency (Figure 3.13). The first prototype assigned one function per robot defined by their attachments. To make the platform more compact, one robotic arm was removed and the distances around all components were reduced. One vacuum head attachment from the removed robot was reassigned to the other two robots, such that each retained its original attachment while also being equipped with an added vacuum suction head (*i.e.* gripper with vacuum and claw with vacuum). This preserved the elimination of cross-contamination by having each robot stack half of the components. Redesigning the component trays were postponed until a later time and the old

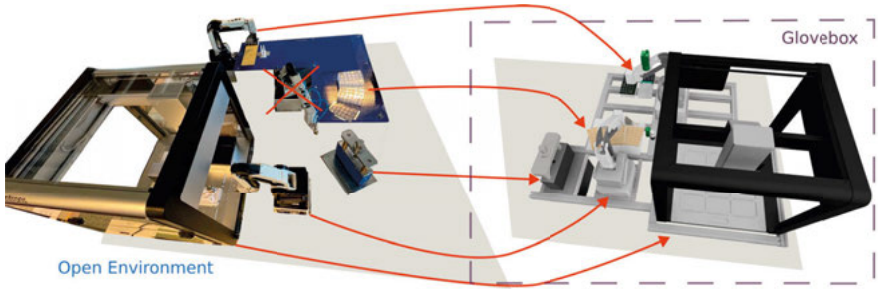


Figure 3.13. Overview of the transition from the first ODACell prototype to ODACell 2.0. Red arrows guide the orientation and replacement of major components during the transition. Grey workbench under each platform estimates the reduced physical footprint of ODACell 2.0.

trays were reused. However, changes in the coin cell setup (addition of an extra spacer) altered the stacking sequence; the adaptable orchestrator programming allowed any stacking configuration to be easily adopted. The addition of two cameras reduced the reliance on strictly hard-coded positions: one positioned above the stacking location, and the other placed offset below the coin cell component holder (Figure 1.1 in green). The bottom camera was used to calculate and correct misalignment after picking up the component, before placing it on the stack.

In upgrading the ODACell platform to include visual capabilities, a couple options were considered. Mounting cameras directly on the robots would have enabled flexible, real-time visual feedback of components, potentially allowing integration with adaptive robotic control algorithms for dynamic adjustment of component placement. However, the camera's bulk would have to be considered in the robotic arm movement and the effects associated with perspective introduced by an angled camera would require calibration. In contrast, cameras fixed to the frame of the platform provided steady images (with the same background and intended imaging subject) that would be easier to apply analytical image techniques to. Flexibility and versatility was sacrificed with choosing this option. Moreover, an extra robot movement position was added to bring the object into the imaging field. Overall, ODACell 2.0 was designed around the specifications of the glovebox (Jacomex PureMod T4-FF, France). Much of the software associated with ODACell 2.0 remained the same.

The focus of the updated ODACell 2.0 was on improved quality control. Solutions to enhance reproducibility, task execution efficiency, and space efficiency motivated the changes and additions to the self-driving lab. The primary challenge associated with operating the platform in an open atmosphere was eliminated with the upgrade. The issue of misalignment persisted despite the camera integration, as shown in benchmarking of Paper III. Although, future improvements will address these following lingering challenges to improve quality control:

- Incorporating a compact infrared spectrometer to aid with electrolyte mixing homogeneity,
- Using high precision optical laser sensors to correct for misalignment complementary to the camera setup,
- Introducing a lid or sealed enclosure to prevent electrolyte evaporation after mixing.

3.3.2 Incorporating statistical model analysis

In Paper III, the electrolyte system shifted from aqueous to nonaqueous. Using more practically relevant electrode materials (NMC811 for high specific capacity and graphite) and electrolyte (LP57), optimization of different addi-

tives were examined. Four additives were chosen, DTD, PES, TMSPi, and VC (chemical structures shown in Figure 1.6) to confirm their individual and combined effects on formation cycling CE. Additional filtering was done using the internal resistance of the cell calculated from the intermittent current interruption technique [36, 37]. Filtering out cells with high resistivity ensured the data added to the Bayesian optimization dataset was all of the same relative quality; samples with electrochemical characteristics inconsistent with what was expected (not captured by visual inspection of the cells) were removed. For the Bayesian optimization, this meant higher quality data, while for the analysis of the results, the internal resistance was an additional metric that would provide insights on the formed SEI.

Baseline metrics without any additives were established prior to optimization. Comparison between the automated assembly and manual assembly, shown in Figure 3.14, set the standard for reproducibility and highlighted the need to further improve the automated assembly process.

Coefficient of variation (CVA), or relative standard deviation, was used here to compare the reproducibility between the manual and automated assembly:

$$\mathrm{CVA} = \frac{\sigma}{\mu}, \qquad \mu = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i-\mu)^2}{n-1}}, \tag{3.1}$$

where μ is the mean, σ is the standard deviation, x_i are the data point values, and n is the number of data points. Each metric (capacity, CE, and internal resistance) produced different CVA; the relationship between capacity and CE is evident from Paper II, sharing similar CVA, but the higher CVA from the internal resistance emphasizes higher sensitivity to both electrochemical processes within the cells as well as to cell assembly processes. Nevertheless, the benchmarking results provided sufficient confidence to proceed with mapping additive effects using a predominantly exploration-driven Bayesian optimization strategy.

The motivation for mapping the effects of electrolyte additives on formation cycling was to increase efficiency of SEI formation, thereby enabling quicker battery deployment. Specifically, a higher CE in the initial cycles indicated reduced Li-loss to SEI formation while still effectively passivating the electrode to prevent further degradation. The additives chosen already indicated alterations to the SEI composition and benefit to the battery lifetime; an expanded upper voltage cutoff of 4.4 V, combined with a voltage hold at this cutoff, was implemented to examine possible activity with the positive electrode to suppress charge loss (*e.g.* from oxygen release or electrolyte oxidation). The exploration relied on Bayesian optimization for experiment planning, although a more refined approach was taken compared to before.

Segmenting the optimization into multiple phases enabled controlled progression through the use of distinct acquisition functions. Compared to adaptive acquisition strategies, explicitly transitioning through optimization phases of-

fered increased flexibility for directing the experimental campaign, at the cost of human intervention. From the simple acquisition behavior observed in Paper II, the domain edges were less frequently sampled compared to within the domain. Model unreliability increased in this region, which was problematic for predicting domain edge behavior. In Paper III, a manual phase was implemented where mostly domain edge samples (single additive in the electrolyte for each additive) at different concentrations were tested and added into the

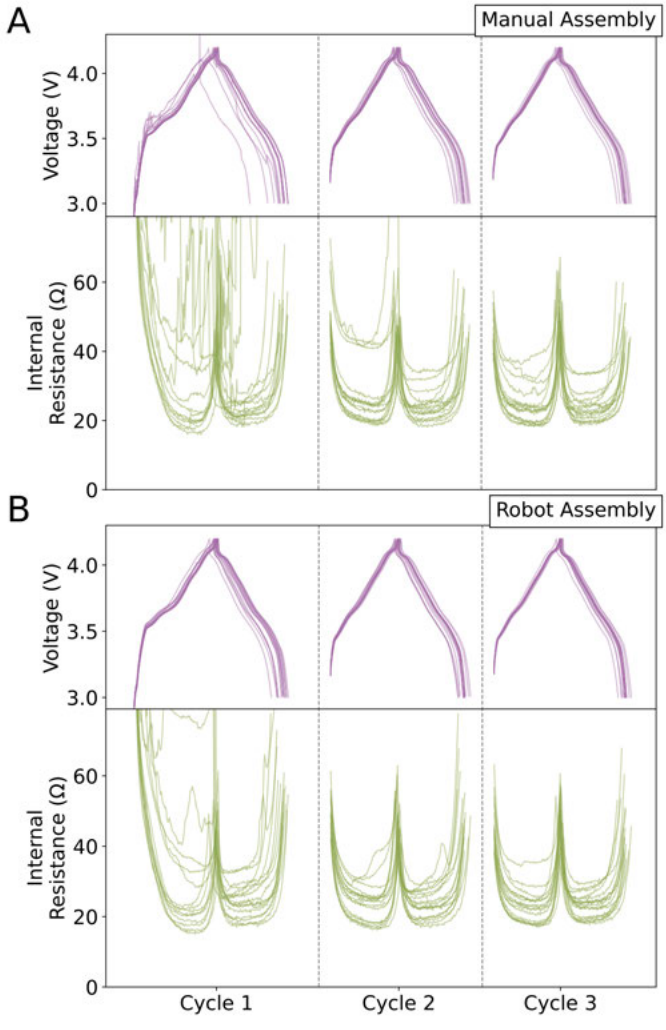


Figure 3.14. Baseline $LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$/graphite with LP57 (1 M $LiPF_6$ in 3:7 v/v ethylene carbonate/ethyl methyl carbonate) full-cell assembly between A) researcher and B) ODACell 2.0 platform, 0.1 C-rate, 3 V-4.2 V voltage cutoffs. Voltage curves (top) and internal resistance of the cell (bottom) are shown as a function of cycle time for at least 15 repeats.

optimization dataset. This strategy attempted to increase global predictability, and produce a more robust surface response.

Unfortunately, the nature of the data rendered the dataset still too limited to be confident in the Gaussian process posterior model. Although noisy data was considered in the Gaussian process, overfitting was still an issue; the exploration-driven sampling resulted in local regions of the parameter space to be sparsely sampled. Consequently, the model was conditioned on very few observations within these regions, leading to high sensitivity to the inclusion or exclusion of individual data points and, therefore, overfitting and having poor cross-validation performance. Instead, bootstrapping ridge regression and random forests were compared against the Gaussian process posterior to determine consistency of trends for the effects of the additives. Both random forest and ridge regression were consistent and provided a better global predictive fit (5-fold cross-validation R^2 for Gaussian process: -0.26, random forest: -0.06, bootstrapped ridge regression: 0.18).

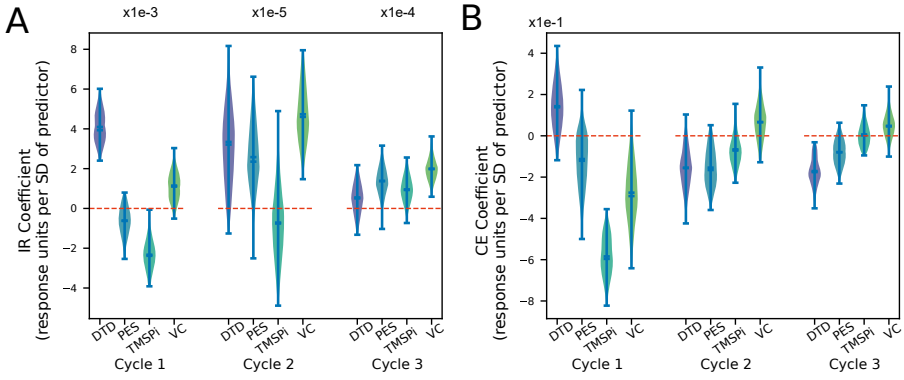


Figure 3.15. Bootstrapped ridge regression coefficients for each additive on A) internal resistance (IR) and B) Coulombic efficiency (CE) across the first three formation cycles. The direction of correlation is determined relative to the red line. The width of the violin reflects the density of bootstrapped models that produced the same coefficient value, thus indicating predictive stability. DTD, 1,3,2-dioxathiolane 2,2-dioxide; PES, prop-1-ene-1,3-sultone; TMSPi, tris(trimethylsilyl) phosphite; VC, vinylene carbonate.

Figure 3.15 shows the effect of each additive on CE and internal resistance for each formation cycle based on the bootstrapped ridge regression fit. The first formation cycle is where the majority of electrochemical reactions occur, and, in the case of a fully formed SEI, the CE in subsequent cycles (cycles 2 and 3) would be expected to remain stable. Consistent with this expectation, cycle 1 exhibited the strongest and most statistically confident correlations; however, electrochemical activity persisted in later cycles. In the first cycle, higher concentrations of DTD predicted higher internal resistance, while higher concentrations of TMSPi predicted lower internal resistance. This correlation disappeared in cycle 2 and 3, where the correlation between VC and internal

resistance became evident. For CE, only TMSPi was negatively correlated in the first cycle (higher TMSPi concentration predicted decreased cycle 1 CE), and DTD was the only additive correlated in cycle 3 (higher DTD concentration predicted decreased cycle 3 CE). The observed trends were in agreement with physicochemical explanations as detailed in Paper III from reported literature.

Non-monotonous additive effects were investigated by observing contour plots of the Gaussian process posterior, keeping in mind the associated error map (Figure 3.16). Only CE of cycle 3 was analyzed as this was the model sequentially built through the Bayesian optimization strategy. The transparency mask clipped CE values below baseline LP57. Regions of high CE were associated with low DTD concentrations consistent with Figure 3.15B. The VC additive was also examined, but provided no substantial differences in the results already observed. The high standard error in many regions (Figure 3.16C) highlights the previous discussion on overfitting and the need for more data.

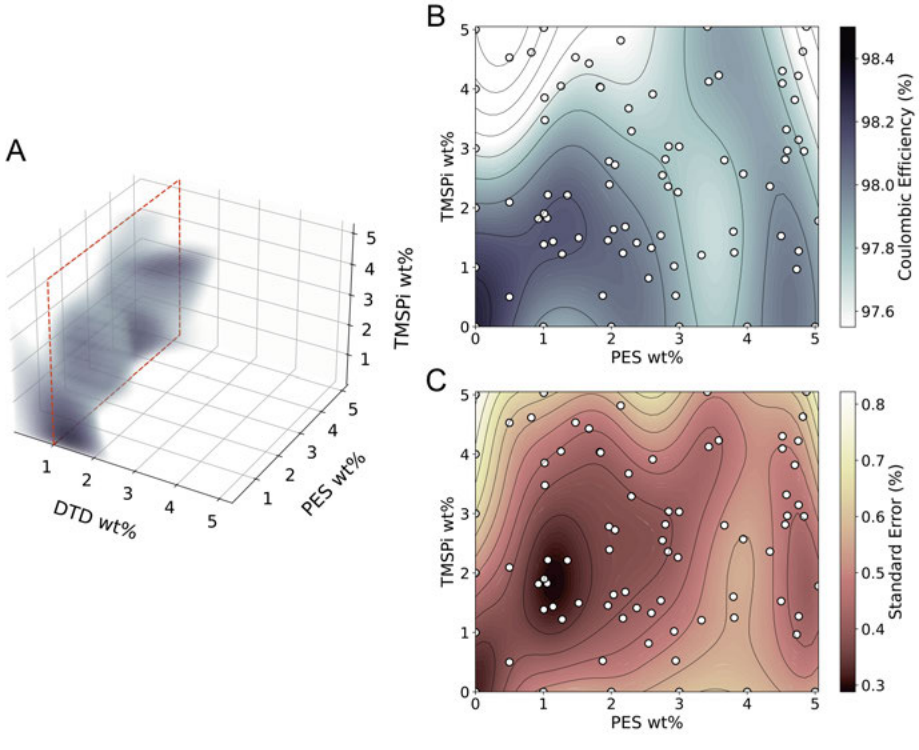


Figure 3.16. Gaussian process posterior for Coulombic efficiency (CE) of the third formation cycle. A) Regions of higher cycle 3 CE compared to baseline are shown in a 3D representation. B) The 2D slice from the red outline in A) shows non-monotonous surface response. A) and B) share the same color map. C) The standard error contour map corresponding to B). DTD, 1,3,2-dioxathiolane 2,2-dioxide; PES, prop-1-ene-1,3-sultone; TMSPi, tris(trimethylsilyl) phosphite.

Longer-term cycling (200 cycles) capacity retention provided another avenue to investigate, shown in Figure 3.17. The third formation cycle CE was uncorrelated to the capacity after 200 cycles. Single additive VC electrolytes retained the highest discharge capacity after 200 cycles compared to the electrolyte blends. Some blends had similar or better capacity retention compared to baseline LP57 electrolyte, but its relation to electrolyte composition requires further investigation. The discrepancy between short-term and long-term metrics to predict lifetime has been a longstanding debate and whether a single global metric for prediction exists is still a topic of discussion [114]. For practical applications, the choice of objective variable(s) and experimental protocols may be as vital as the optimization setup itself or hyperparameter tuning.

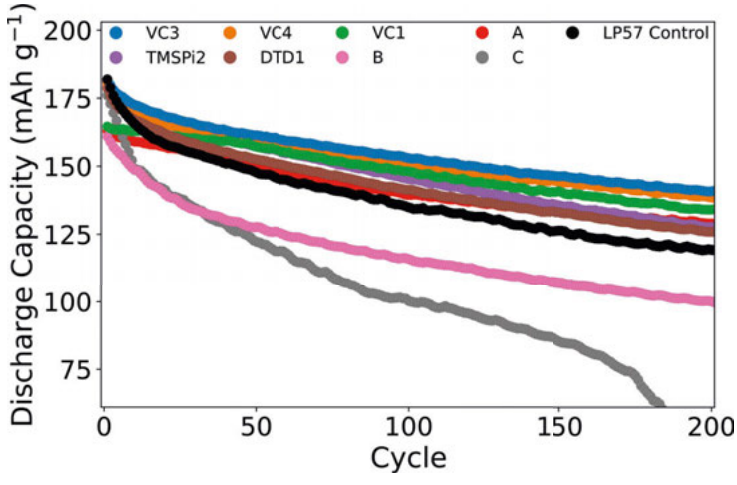


Figure 3.17. Capacity retention for $LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$/graphite coin cells during 200 cycles (0.5 C-rate, 3.0 V-4.4 V). Single additive electrolytes (denoted by VC, TMSPi, or DTD, followed by weight percentage concentration) along with the electrolyte blend A (DTD 0.50 wt% + PES 0.93 wt% + TMSPi 1.82 wt% + VC 3.06 wt%) retained higher capacity than the LP57 control (1 M $LiPF_6$ in 3:7 v/v ethylene carbonate:ethyl methyl carbonate); electrolyte blends B (DTD 1.87 wt% + PES 1.47 wt% + TMSPi 4.53 wt% + VC 0.84 wt%) and C (DTD 1.72 wt% + PES 4.76 wt% + TMSPi 1.27 wt% + VC 0.69 wt%) achieved lower capacity retention than the control. DTD, 1,3,2-dioxathiolane 2,2-dioxide; PES, prop-1-ene-1,3-sultone; TMSPi, tris(trimethylsilyl) phosphite; VC, vinylene carbonate.

In this thesis, across all the studies, the surrogate model for Bayesian optimization was fixed to a Gaussian process model and no alternative models (or change in kernel) were considered. The focus has been predominantly on the self-driving lab and methodology. Different uses of Bayesian inference and experimental planning algorithms exist [21]. A more in-depth investigation of the analytical framework is expected to reduce discrepancies and enhance the statistical robustness of the conclusions. The ODACell platform supports

adaptable experimental planning, even though data analysis in the Paper III was carried out offline.

3.4 Broader application: zinc

Paper IV applied the self-driving lab and optimization workflow to zinc metal batteries. Although the workflow and hardware remained unchanged, the new electrode and electrolyte materials necessitated some setup reassessment and recalibration. Liquid handling reverted back to aqueous solutions making liquid handling easier than nonaqueous solutions. The Cu||Zn asymmetric (half-cell) configuration used copper and zinc foils to investigate zinc plating

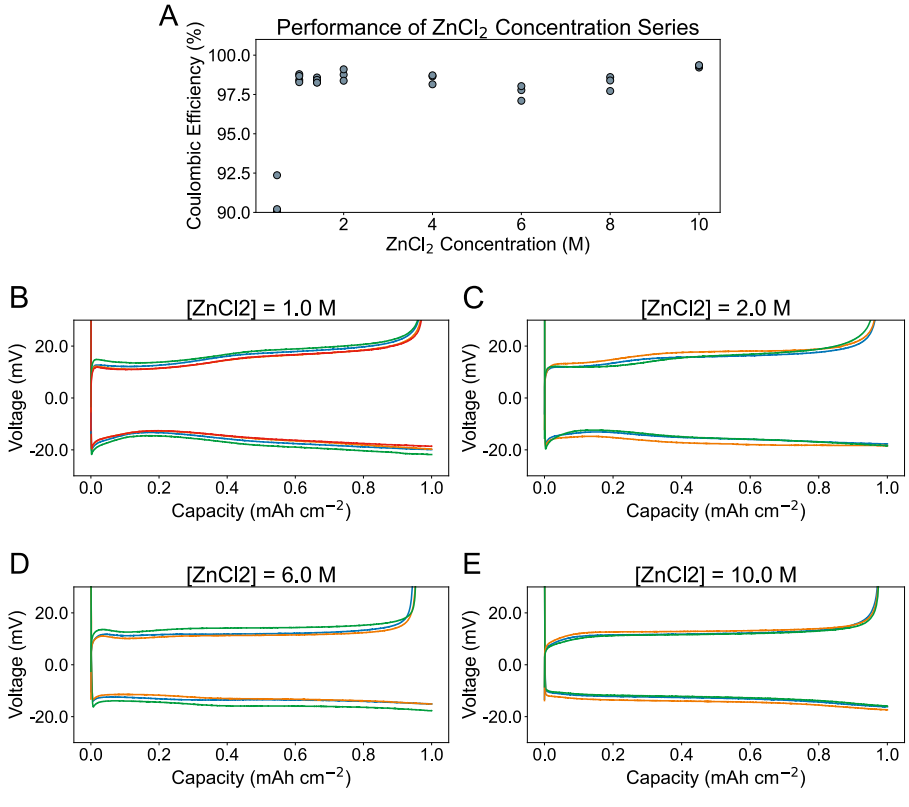


Figure 3.18. Reproducibility and cycling polarization of Cu||Zn cells for different concentrations of $ZnCl_2$ electrolyte. A) Mean Coulombic efficiency of cycles 14, 15, and 16 for different concentrations of $ZnCl_2$ using 1.0 mA cm^{-2} for 1.0 mAh cm^{-2}, indicating minor differences observed across concentrations. At least three repeats were conducted for each concentration step. Zinc plating and striping polarization curves for B) 1.0 M, C) 2.0 M, D) 6.0 M, and E) 10.0 M $ZnCl_2$, respectively (0.5 V stripping cutoff). Repeats are shown in different colored pairs, indicating high reproducibility.

and stripping on copper foil. Unlike the coated active material composite for the lithium intercalation material, the typical copper foil thickness (<15 μm) was incompatible with the setup. Because the vacuum suction strength was not adjustable, the thin copper foil experienced wrinkling during pickup, so a slightly thicker foil (20 μm thickness) was used instead. Compared to the composite electrodes, the foil electrodes exhibited lower porosity and mass, which led to increased misplacement in the assembly process due to static electricity.

Different $ZnCl_2$ salt concentrations were first tested to determine reproducibility as well as the domain bound for mapping. $ZnCl_2$, similar to LiTFSI, has a high solubility limit in water enabling a water-in-salt electrolyte. At least three repeats for various $ZnCl_2$ concentrations yielded reproducible CE and voltage curves, shown in Figure 3.18. The optimization bounds for the other additives were decided through literature reports on each additive, and up to 2.0 M $ZnCl_2$ was decided as there seemed minor improvements above this concentration.

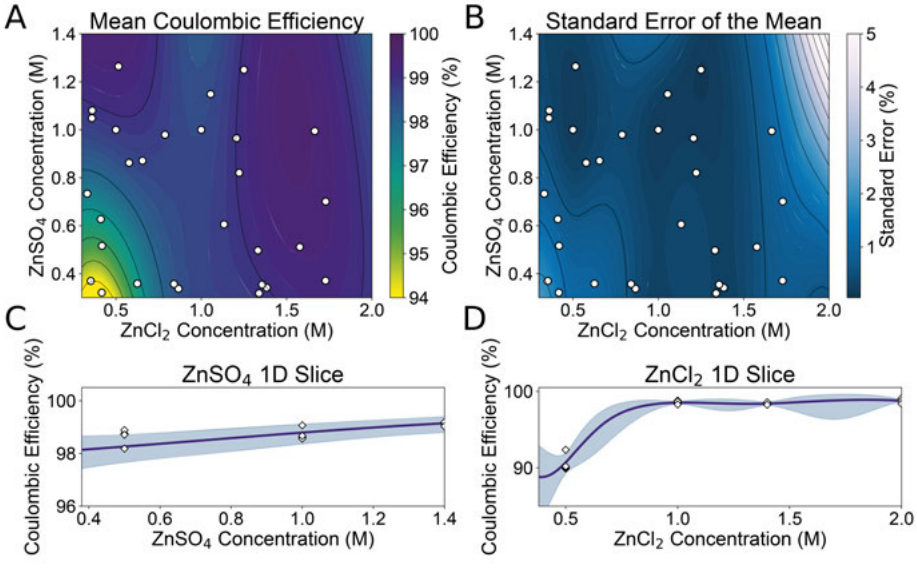


Figure 3.19. Gaussian Process posterior after Bayesian optimization shows higher Coulombic efficiency for high salt concentrations. A) Contour map of the objective (Coulombic efficiency) for the two zinc conducting salts ($ZnSO_4$ and $ZnCl_2$) when removing the additives (setting concentration of thiourea, ammonium phosphate monobasic, and sodium dodecyl sulfate to 0.0 M). Experimental data used to train the model are flattened and overlaid on the contour map. B) The model standard error of the corresponding contour map in A). C) Posterior mean and 95% credible interval (shaded) for $ZnSO_4$, removing all other parameters. Experimental data for 0.5 M, 1.0 M, and 1.4 M $ZnSO_4$ are shown. D) Posterior mean and 95% credible interval (shaded) for $ZnCl_2$, removing all other parameters. Experimental validation data for 0.5 M, 1.0 M, 1.4 M, and 2.0 M $ZnCl_2$ are shown.

The Bayesian optimization compositional design space consisted of three additives along with two zinc salts. The resulting Gaussian process posterior was examined for the salts and additives separately. Virtually removing the additives, the effect of combined zinc salts, shown in Figure 3.19, seemed to provide no benefit over single salt electrolytes. The low CE observed for $ZnCl_2$ concentrations below 1.0 M created a local region of low CE (<96%). The corresponding standard error map indicated uncertainty in this region (relative to the immediate surroundings), suggesting that the magnitude and rate of the CE decline were not yet fully resolved; the four electrolytes containing 0.5 M $ZnSO_4$ did not exhibit a notable decrease in CE compared to the other $ZnSO_4$ concentrations. With similar challenges from the limited dataset, ridge regression was also used to separate the influences of each additive on the CE.

Regardless of the salt concentrations, TU decreased the CE substantially in all electrolyte blends. The effects of MAP and SDS were largely insignificant in comparison. This was seen evidently in the Gaussian process posterior as well as after fitting the data using ridge regression (Figure 3.20). The MAP additive was the only component to show a positive correlation with CE, although the low coefficient value indicating weak correlation necessitates additional investigation.

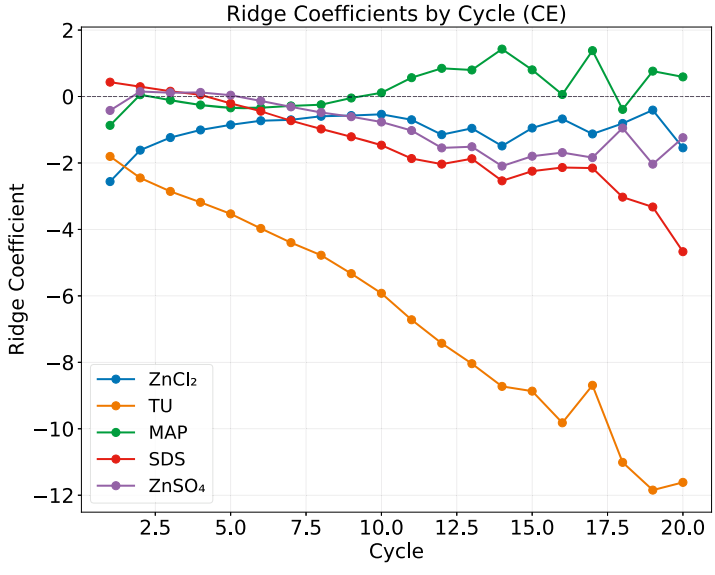


Figure 3.20. Ridge regression feature coefficients for Coulombic efficiency (CE) across 20 cycles. MAP, ammonium phosphate monobasic; TU, thiourea; SDS, sodium dodecyl sulfate.

The application of Bayesian optimization with Gaussian processes and statistical analysis can be compared and contrasted across the lithium and zinc battery systems studied in this thesis. The dataset for Paper III comprises over

100 data points (including replicates), whereas Paper IV contains approximately 80. Although Paper IV explores a higher-dimensional design space (five variables compared to four in Paper III), cross-validation and observed trends appear more robust in the zinc study. The interplay between the dimensionality, dataset size, and response requires discussion. In this case, the domain and associated effect is critical. Although the concentrations range from 0 wt% to 5 wt% for the lithium study (compared to the smaller domain range in the zinc study), the resulting impact on CE was minimal. The subtlety of these effects relative to experimental noise necessitated a larger dataset to resolve marginal differences. Whereas, the zinc study exhibited a pronounced response, mainly from the TU additive, which led to stronger predictive model performance. However, because of that, the insight into the individual roles of the other additives (MAP and SDS) may potentially be limited. Meaningful conclusions and the optimization or mapping progression seem dependent on these experimental or system-specific factors. In Paper IV, another advanced characterization technique was used to confirm and provide an explanation for the significant decrease in CE with the additive TU.

3.4.1 Advanced characterization: electrochemical quartz crystal microbalance with dissipation monitoring

The interphase properties were investigated with and without the TU additive present in the control electrolyte (1.2 M $ZnCl_2$ + 1.0 M $ZnSO_4$). The duration of plating and stripping cycles was shortened for these experiments, although the cycling curves remained relatively consistent with the coin cell results (one minute compared to one hour half-cycles in the coin cells). Nevertheless, immediate differences can be seen when including TU, shown in Figure 3.21. The larger frequency and dissipation shifts with TU indicate its ability to alter the structural evolution of the zinc deposition. The steady sloping envelop in the frequency shift (Figure 3.21A) without TU reveals the gradual buildup of a passivation layer over the testing time; in contrast, mass accumulation occurs more rapidly in the presence of TU. Additionally, the sharp increase in dissipation shift (Figure 3.21) during the initial stripping phases with TU along with a sharp increase to high mass per mole of electron (Figure 3.21) could suggest the formation of a porous network with liquid-filled pores; in later cycles, deposition of zinc in these pores mask additional mass changes and the mass per mole of electron decreases away from the expected ideal divalent zinc plating and stripping mass per mole of electron value (32.69 g mol^{-1} e^{-1}).

Complementary data from advanced characterization reveal the physicochemical mechanisms underlying the CE trends observed with ODACell. As with the gas analysis in Paper II, ODACell provides the foundation for integrating advanced characterization methods after initial rapid screening.

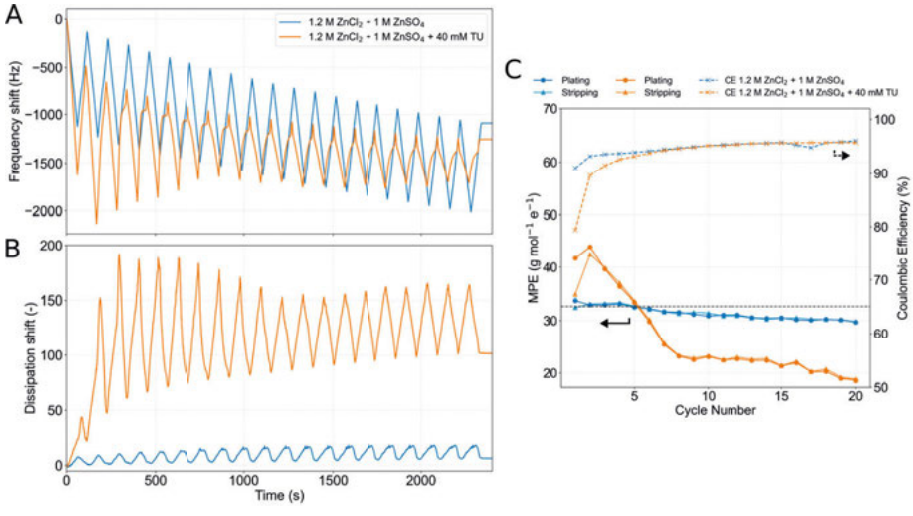


Figure 3.21. Electrochemical quartz crystal microbalance with dissipation monitoring for zinc plating and stripping on copper in the presence and absence of thiourea (TU). A) frequency and B) dissipation shifts of the third overtone highlight anomalies to the passivation layer. C) mass per mole of electron (MPE) values show expected MPE without TU (32.69 g mol^{-1} e^{-1}), but deviations in the presence of TU. Cycling was done at 1.0 mA cm^{-2} plating/stripping current density for 0.0167 mAh cm^{-2}/0.5 V cutoff, respectively.

4. Concluding remarks and outlook

This doctoral thesis details the development, evolution, and application of a self-driving lab for liquid electrolyte formulation and evaluation in research-scale CR2025 battery coin cells. The contributions can be grouped into three categories: the hardware for the self-driving lab, the Bayesian optimization workflow for the self-driving lab, and the insights regarding the chemistry for each model system,.

The hardware and the design of the automated platform created the skeleton on which this self-driving lab's functions and capabilities were defined. The accompanying software, built around the platform, were constantly improved upon and refined. The upgraded design shown in Paper II and onward underscores the flexibility of the setup and the possibility to improve further with the inclusion of better positioning aids and additional attachments for the robots or the platform as a whole. The design incorporates the possibility to add modular instruments for advanced characteristics. While using the self-driving lab for diverse applications, more challenges presented itself creating an exciting and ever-changing engineering marvel.

The implementation and choice of optimization schemes can be very diverse. Three different optimization strategies were used to guide experimentation in this thesis. A single phase (with an unchanging acquisition function) provides the simplest solution, and for Paper II, an adaptive acquisition function increased sampling diversity by penalizing candidates close to already-evaluated or high-performing points. Paper IV separated the optimization into an exploration phase followed by a manual phase where domain edge data points were added to the dataset. Paper III went beyond this and used three phases, switching from an exploration acquisition function to an exploitation one, again supplemented by manual points. Choosing when to switch phases and what scheme to follow (or how many phases) was largely based on my intuition on the optimization progression and the available resources. Although the experimental planning became more refined with each application, use of different models, and statistical analysis was conducted after the dataset was collected. Therefore, there are still improvements to be gained in enhancing the self-driving lab's machine intelligence for active learning.

Various battery chemistries were used as model systems to demonstrate the versatility of the self-driving lab and reveal insights for each system. Paper I and II probed the detrimental effects of water in the electrolyte on CE and discharge capacity through short-term cycling. In an open laboratory atmosphere, the hygroscopic electrolyte hindered reproducibility, leading to significant data

scatter. Observable performance differences were negligible for sufficiently low water content; however, under the nitrogen atmosphere in Paper II, and with the co-solvents TMP and/or DMSO, even minor water contamination exhibited non-negligible HER, and the combination of ACN and water expedited the reaction rate.

In the nonaqueous NMC811||graphite system, additives for enhancing efficiency of formation cycling was investigated instead of degradation. Although the objective variable remained unchanged, the interpretation of the effects seen were inverted. In Paper III, a new challenge associated with the increased baseline CE emerged creating difficulty in being able to perceive significant changes to objective. Thus, the introduction of internal resistance aided in ensuring high-quality data and supporting the insight drawn from statistical analysis. Ultimately, DTD and TMSPi additives showed the most prominent effects on formation cycling, which did not completely translate to longer lifetime. The complicated interplay between short-term and long-term performance metrics yields an interesting outlook for further exploration; the impact of testing protocol, chemistry, and cell format may be significant.

The ODACell 2.0 platform demonstrated compatibility with battery systems beyond lithium, namely zinc metal batteries (Paper IV). With the copper and zinc foil electrodes, the CE again took on a slightly different physical interpretation: the zinc plating-stripping efficiency on the copper substrate. Then, the additives were tested for their ability to modify zinc deposits or surface interactions to adjust HER kinetics. Paper IV showed that the TU additive caused a pronounced decrease in CE across all electrolyte formulations. This dominant effect overshadowed those of the other additives, enabling its rapid identification despite the smallest dataset. The chemical findings for each system and their additives are summarized in Table 4.1.

Table 4.1. *Summary of findings for the effects of different additives optimized for each system studied.*

System	Base Electrolyte	Positive Effect Additives	Neutral Effect Additives	Negative Effect Additives
Paper I: LFP||LTO	DMSO ($LiClO_4$)	-	-	H_2O
Paper II: LFP||LTO	1 m salt ($LiClO_4$ + LiTFSI)	DMSO, TMP	-	H_2O, ACN
Paper III: NMC811||Graphite	LP57	TMSPi, DTD, VC	PES	-
Paper IV: Cu||Zn	H_2O ($ZnSO_4$)	MAP	SDS	TU

5. Populärvetenskaplig sammanfattning

I takt med att vi övergår från fossila bränslen till förnybar energi blir effektiv energilagring allt viktigare. Batterier är en viktig lösning, men den växande energiefterfrågan kräver att vi utvecklar batterier som håller längre, är säkrare och har bättre prestanda. Inuti ett batteri kan prestandan förbättras genom att modifiera tre huvudkomponenter: den positiva elektroden, den negativa elektroden och elektrolyten mellan dem. Elektrolyten, som vanligtvis är en vätska, kan tillverkas av många olika kemiska ingredienser. Det stora antalet möjliga kombinationer gör testningen svår, både fysiskt och när det gäller att bestämma vilka kombinationer som ska undersökas. Dessutom är det tidskrävande att testa batterier, så det är viktigt att påskynda forskningen inom detta område för att uppnå framtida energimål.

I denna avhandling utvecklade jag ett "självkörande laboratorium" kallat ODACell för att ta itu med för att ta itu med de utmaningar som följer av elektrolytens stora designutrymme. Detta system möjliggör reproducerbar batteritillverkning och testning, samtidigt som det på ett intelligent sätt utforskar elektrolytblandningar för att maximera informationsvinsten med minimalt antal experiment. Det självkörande laboratoriet består av två huvuddelar: hårdvara och mjukvara.

För hårdvaran designade jag en robotplattform med robotarmar och en pipetteringsrobot som är fastmonterad. Specialdesignade brickor och tillbehör håller batterikomponenterna, vilket gör att systemet kan montera batterier automatiskt. Inledningsvis hanterade tre robotarmar högspecialiserade uppgifter: att plocka upp komponenter, flytta komponentstapeln till olika platser och placera det tillverkade batteriet i batteritestkanaler. Förbättringar av plattformen eliminerade dock en robotarm och ökade utrymmeseffektiviteten. Pipetteringsroboten förbereder exakta elektrolytformuleringar och doserar elektrolyten i batteriet. Allt detta styrs och koordineras med hjälp av Python-programmering.

För programvaran använde jag en optimeringsalgoritm som kallas Bayesiansk optimering. Denna algoritm lär sig av experimentella data och modellerar hur olika elektrolytkompositioner påverkar batteriets prestanda. Den föreslår sedan de bästa följande experimenten (antingen för att förbättra prestandan eller för att minska osäkerheten) och styr utforskningen på ett effektivt sätt.

Tillsammans bildar dessa komponenter en sluten loop: algoritmen föreslår en elektrolyt, batterier byggs och testas med motsvarande elektrolyt, och resultaten analyseras automatiskt för att informera nästa experiment. ODACell kan köras kontinuerligt med minimal mänsklig inblandning.

Hårdvaran i ODACell testas först i Artikel I. Utan strikt kontroll av miljön under montering försämrades reproducerbarheten trots batterier med samma

elektrolyt och andra komponenter. Den uppgraderade ODACell blev mer utrymmeseffektiv och kunde passa in i en kontrollerad atmosfär. Detta möjliggjorde bredare testning av olika elektrolytsystem. Efter uppgraderingen av ODACell (och införlivandet av den maskinintelligenta experimentplaneraren) demonstrerar jag dess tillämpning på tre olika batterielektrolytsystem. Först undersökte jag vattenhaltiga elektrolyter för litiumjonbatterier i Artikel II, som vanligtvis fungerar under strikt vattenfria förhållanden. Detta beror på att vissa kemikalier reagerar med vatten och bildar gaser och frätande ämnen som förstör batteriet. Med hjälp av ODACell identifierades lösningsmedelsblandningar som fungerade bra med vatten och man kvantifierade hur mycket vatten systemet kunde tolerera. För det andra kartlade ODACell i Artikel III effekterna av prestandahöjande tillsatser i konventionella icke-vattenhaltiga litiumjonbatterier. Av fyra testade tillsatser visade tre positiva effekter, var och en av olika skäl, vilket avslöjade hur små förändringar i sammansättningen kan förbättra batteriets prestanda. Slutligen, i Artikel IV, tillämpade jag ODACell på en nästa generations batterikemi: zinkbatterier. Till skillnad från litium är zink rikligt förekommande, säkert och kostnadseffektivt, men dess elektrolyter medför olika utmaningar. ODACell undersökte effektivt fem elektrolytkomponenter och identifierade snabbt en som försämrade prestandan avsevärt—ett resultat som senare bekräftades med avancerade karaktäriseringstekniker.

Även om ODACell inte är en färdig produkt lägger detta arbete grunden för accelererad batteriforskning. Jag beskriver flera förbättringar—såsom integrering av nya sensorer och ökad reproducerbar elektrolytberedning—som skulle kunna förbättra framtida system. ODACell visar hur automatisering och maskininlärning kan kombineras för att utforska komplexa material snabbare och mer systematiskt, vilket öppnar dörren för forskare att anpassa denna metod till andra vetenskapliga utmaningar.

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